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## SURVEY

# Building Trustworthy AI in Healthcare

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**ABSTRACT** Artificial Intelligence (AI) has introduced significant innovations in healthcare, yet its adoption has been slower compared to other industries. A key barrier lies in concerns over trustworthiness, as healthcare stakeholders often express skepticism stemming from the opacity of AI systems, risks of privacy breaches, potential biases that exacerbate social inequities, and unresolved safety issues. Ensuring trustworthy AI is therefore essential for its safe, effective, and responsible integration into clinical practice. This integrative review examines four fundamental dimensions of trustworthy AI in healthcare: transparency and explainability, privacy and security, robustness and safety, and bias and fairness. Unlike prior surveys that either addressed trustworthiness in AI broadly or focused narrowly on traditional machine learning and deep learning (DL), this work extends the discussion to Large Language Models and Agentic AI, both of which are poised to play transformative roles in healthcare. We also explore the intersections among trustworthiness dimensions and highlight the often-overlooked roles of user trust and organizational accountability in shaping real-world adoption. Finally, we provide targeted recommendations, outlining practical tools, frameworks, and strategies tailored to different model classes to address concrete trustworthiness challenges in healthcare AI.

**INDEX TERMS** Trustworthy AI, trustworthy LLM, bias, fairness, privacy, security, robustness, safety, explainability, interpretability, transparency, healthcare, agentic AI.

## I. INTRODUCTION

### A. BACKGROUND

Artificial Intelligence (AI) is a science that focuses on building systems with the potential to reason, learn, simulate, and even extend or exceed human intelligence [1]. AI can be trained to produce different results, ranging from predictions, multimodal content, recommendations, summarization, and certain real-life decisions with real consequences.

Before the advent of Foundation Models (FMs), discussions on AI mostly focused on traditional machine learning and deep learning techniques. These techniques are commonly applied to tasks like Natural Language Processing (NLP), prediction, anomaly detection, recommendation systems, and computer vision, to mention a few. However, the emergence of foundation models, especially generative AI (GenAI), has significantly changed the landscape. Large

Language Models (LLM), for instance, have attracted widespread attention and accelerated the adoption curve of AI in healthcare and other similar safety critical domains like autonomous vehicles.

### B. AI IN HEALTHCARE

In modern healthcare, AI is transforming diagnosis, treatment, and disease management, building on early expert systems like MYCIN [2], which diagnosed bacterial infections, and CADUCEUS [3], which replicated clinical reasoning. In addition to these early expert systems, the evolution of medical AI progressed through intermediate statistical and symbolic approaches, most notably decision-tree systems [4], [5] and Bayesian networks [6], [7], which provided probabilistic reasoning, interpretable decision pathways, and early risk-prediction capabilities that shaped modern machine learning methods [8], [9]. Current machine learning and deep neural network models enhance precision medicine, diagnostic accuracy, and patient outcome predictions. Notable

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advancements include AlphaFold [10], which resolved the protein folding problem, and In Silico Trials [11], [12], which simulate drug discovery, leading to faster diagnoses, personalized care, and potential economic savings [13]. Furthermore, [14] demonstrated measurable economic value and positive clinical impacts of AI across diverse healthcare settings.

In addition, Natural Language Processing (NLP) has optimized the use of Electronic Health Records (EHRs) by processing unstructured clinical data. AI applications span various medical fields: for instance, in cardiology, it aids early detection and personalized treatment; in ophthalmology, it diagnoses eye conditions from imaging; in dermatology, Convolutional Neural Networks (CNN) assist in skin cancer detection; in neurology, it diagnoses disorders and tailors therapies; in radiology and cancer treatment, it improves image interpretation and personalizes protocols; and in emergency medicine, it optimizes patient triage and resource allocation. These innovations summarized in Table 13, apply deep learning, computer vision, GenAI, NLP and other techniques in precision medicine, drug discovery, and medical imaging, viewing AI as a powerful tool to augment human capabilities in diagnosis, testing, treatment, monitoring, and long-term care [15], [16].

Recently, agentic systems have shown promise in reducing administrative burdens in strained healthcare systems, with pilot programs in the UK's National Health Service (NHS) providing clear examples [17], [18]. By automating routine tasks like scheduling, data entry, and patient triage, these systems can free up clinicians' time, allowing them to focus on direct patient care. However, the use of agentic systems in healthcare is not without challenges. Because these systems handle sensitive data and directly influence clinical workflows, their trustworthiness is paramount. This necessitates rigorous evaluation of their transparency, reliability, and fairness, as well as continuous monitoring to prevent errors or biases from compromising patient safety and public confidence.

### C. SLOW ADOPTION AND MOTIVATION

Despite these potential benefits, the adoption of AI in healthcare, which is dependent on its trustworthiness, is slower than in other domains [19], [20], [21], [22], [23]. Several works have identified various key barriers hindering the broad adoption of clinically-validated AI applications. For instance, healthcare professionals are reluctant to adopt large-scale AI-based medical diagnosis support systems mainly due to their lack of interpretability and transparency [19], [20], [21], [24], [25]. Reference [15] argues that closed box models and algorithmic bias erode trust among healthcare providers and the public, thus limiting the broader adoption of AI-driven healthcare systems. Meanwhile, the adoption of LLM specifically in healthcare has been hampered by concerns about factual inaccuracy, incoherence, and the risk of hallucinations [23].

According to [21], many digital health tools, especially Computer Decision-Support Systems (CDSS), often do not make it beyond pilot phases due to opacity of deep learning. Other barriers include fragmented data, privacy restrictions, inadequate verifiability, the lack of standardized medical records, the strict legal and ethical constraints necessary to protect patient privacy, regulatory approval, financial incentives, data quality and integration, and clinician education, among others [13], [19]. To better understand these challenges, we can classify them into a structured barrier taxonomy: technical barriers (e.g., data fragmentation, verifiability gaps, and distributional shift), organizational barriers (e.g., misaligned and financial incentives, inconsistent data standards, and limited regulatory readiness), and human-factor barriers (e.g., lack of clinician education, trust calibration and interpretability needs). This structure makes explicit which trust dimensions each barrier most directly impacts.

The attributes consistently recognized as critical for the adoption of AI among healthcare providers, professionals, and patients include transparency, explainability, accountability, reliability, safety, fairness, and the ability to engender trust. Moreover, [20] demonstrated that potential users often hesitate to adopt AI due to a lack of measurable indicators of trustworthiness.

### D. TRUSTWORTHY AI

Trustworthy AI (TAI) refers to AI systems designed for reliable, ethical, and transparent operation, aligning with human values and societal expectations, engendering user trust, and preventing harm [1], [20], [26]. To clarify the conceptual boundaries used throughout this integrative review, we differentiate among three related frameworks. TAI, the main focus of this work, specifically emphasizes technical aspects with a focus on reliability, transparency, and accountability [1], [20], [26]. This is distinct from Responsible AI, which focuses on the ethical and organizational practices of AI governance, such as fairness, transparency, accountability, and privacy. Finally, Safe AI addresses the broader societal risks posed by AI systems [27]. Although these concepts have different focal points such as technical, organizational, and societal, respectively, they share significant overlaps and are all integral components of comprehensive AI governance [27].

### E. THE NEED FOR TAI IN HEALTHCARE

AI in healthcare raises significant data privacy concerns, as training data often contains sensitive personal information, making breaches or misuse highly consequential for the stakeholders [13]. The high-stakes nature of healthcare demands that AI be trustworthy, addressing risks such as data breaches or misuse, hallucinations, bias, security vulnerabilities, and unpredictable behavior while ensuring strict data privacy under regulations like GDPR and Health Insurance Portability and Accountability Act (HIPAA). Trust,

transparency, and explainability are essential for adoption and building confidence with both clinicians and patients, requiring systems that are interpretable, accountable, and complementing, rather than replacing human judgment [25], [28], [29], [30], [31]. Although there are models and trust frameworks that aim at improving user confidence and system transparency, there is still a gap in systematic evaluation of user acceptance and real-world TAI impact [26], [32]. AI aims to alleviate clinician burden, improve diagnostics, and streamline workflows for better patient outcomes. However, if untrusted, AI's inherent value is undermined, or better put - an AI system, regardless of its advancement, holds no value if the very people it aims to help cannot or will not use it.

Global efforts prioritize trustworthy AI, with numerous frameworks emphasizing Bias & Fairness, Privacy & Security, Robustness & Safety, and Explainability, Interpretability & Transparency (EIT) as core pillars [20]. Examples include the European Union's (EU) General Data Protection Regulation (GDPR), AI Act, Digital Services Act (DSA), and Digital Markets Act (DMA), promoting transparency, safety, accountability, and fairness. The High-Level Expert Group on AI (HLEG) framework also outlines seven widely adopted TAI dimensions, including fairness, transparency, robustness, and privacy [25], [33], [34]. In the US, the National Institute of Standards and Technology's (NIST) AI Risk Management Framework (RMF), AI Bill of Rights, and Algorithmic Accountability Act provide guidance, whereas Defense Advanced Research Project Agency's (DARPA) Explainable AI (XAI) program and Government Accountability Office's (GAO) frameworks reinforce responsible AI. Canada's Consumer Privacy Protection Act (COPA) and the defunct AI and Data Act (AIDA) embed fairness, transparency, and safety into national AI governance. The Organization for Economic Cooperation and Development (OECD) is a foundational source for TAI frameworks, defining key dimensions and promoting ethical AI in healthcare through patient-centric guidelines and clinical validation [25], [33]. Healthcare-specific laws like HIPAA mandate strict privacy protections, highlighting the need for safeguards like encryption and anonymization, although balancing these with explainability remains a challenge [1], [13], [15], [25], [34], [35], [36], [37].

Across the multiple frameworks surveyed, common pillars such as fairness, transparency, privacy, robustness, and accountability recur, but the frameworks differ in emphasis and operationalization. European instruments are ethics- and rights-driven, U.S. guidance is measurement- and risk-management-oriented, international bodies emphasize equity and public benefit, and engineering standards focus on process integrity and system safety. These inconsistencies create practical challenges for healthcare AI, where sensitive data, high-stakes decisions, and clinical validation demands require metrics, such as calibration, subgroup fairness, privacy budgets, and real-world safety indicators,

that are more stringent and domain-specific than those in general-purpose AI frameworks. These differences manifest in distinct evaluation priorities, such as autonomy versus safety; however, patient safety serves as the critical unifying metric that must bridge these regulatory inconsistencies to ensure practical deployment.

Beyond these regional and national policies, several crucial domain-specific and international guidelines are essential. The World Health Organization's (WHO) 2021 Guidance on Ethics and Governance of AI for Health establishes six core principles for the public benefit, including protecting autonomy and ensuring inclusivity [38]. Foundational definitions are provided by the International Medical Device Regulators Forum (IMDRF), which created the Software as a Medical Device (SaMD) category and its risk-based classification framework [39]. On the regulatory front, guidelines from the U.S. Food and Drug Administration (FDA), Health Canada, and the European Medicines Agency (EMA) for AI/ML-based SaMD are critical. The FDA's framework is notable for its predetermined change control plan, an innovative pathway for adaptive algorithms to learn from real-world data without requiring new clearance for every change [40]. Similarly, the European Medicines Agency (EMA) emphasizes a lifecycle approach, requiring that AI/ML models are robustly validated and monitored from development through the post-authorization phase to ensure patient safety [41]. Furthermore, Health Canada uses the IMDRF's risk-based classification (from Class I to IV) to determine the level of evidence required for a SaMD's safety and effectiveness [42].

From a practical engineering standpoint, the IEEE P7000 series provides a framework for Ethically Aligned Design (EAD), with specific standards like P7001 on transparency, P7002 on data privacy and P7003 on algorithmic bias [43], [44], [45]. This is complemented by international standards like IEC 62304 (defining the software development lifecycle and risk management process), which is essential for ensuring and demonstrating product safety and reliability [46].

The ethical principles from UNESCO guidelines ensure responsible AI design, improving the trustworthiness of digital health systems. These collective efforts justify the focus on bias & fairness for equitable outcomes, privacy & security for rights protection, robustness & safety for dependable behavior, and explainability & interpretability for transparency and accountability. A quick reference summary of these essential regional, national, and international frameworks is provided in Table 1.

## F. LITERATURE SEARCH STRATEGY

A comprehensive literature search was conducted to systematically identify, synthesize, and evaluate literature on trustworthy AI and LLMs in healthcare. The initial search spanned the period from 2017 up to 2025 and utilized the primary databases Scopus and Google Scholar. The strategy incorporated both peer-reviewed sources (journals,

TABLE 1. Quick-Reference governance and regulatory frameworks for TAI in healthcare.

| Framework/Regulation | Geographic Scope    | Primary Focus                            | Key Mandates for Healthcare AI                                                                                                        |
|----------------------|---------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| EU AI Act            | European Union (EU) | Risk-based regulation of AI systems      | Classifies most healthcare AI as high-risk, mandating stringent requirements for data, transparency, human oversight, and safety.     |
| GDPR                 | European Union (EU) | Data privacy and protection              | Provides structured guidance on the collection, processing, and use of sensitive personal (health) data.                              |
| NIST AI RMF          | United States (US)  | Risk management guidance                 | A voluntary framework to help organizations identify, measure, and manage AI risks to reinforce responsible practices.                |
| HIPAA                | United States (US)  | Health data privacy and security         | Mandates strict privacy protections and security safeguards for all Protected Health Information (PHI).                               |
| CPPIA/AIDA (defunct) | Canada              | National AI governance & data privacy    | AIDA (part of Bill C-27) regulates high-impact AI systems, while CPPIA modernizes data privacy rules. Both embed fairness and safety. |
| WHO Guidance         | Global              | Principles-based ethical guidance        | Establishes six core principles for global AI health policy, including protecting autonomy and ensuring inclusivity.                  |
| FDA/EMA SaMD         | US / EU             | Medical device regulation                | Governs how AI/ML models are validated and approved. Introduced PCCP for adaptive algorithms.                                         |
| IEEE P7000 Series    | Global              | Engineering standards (Ethics by Design) | Provides actionable processes for Ethically Aligned Design, e.g., P7001 (Transparency) and P7003 (Algorithmic Bias).                  |

Note: AIDA: Artificial Intelligence and Data Act; CPPIA: Consumer Privacy Protection Act; EMA: European Medicines Agency; EU: European Union; FDA: Food and Drug Administration; GDPR: General Data Protection Regulation; HIPAA: Health Insurance Portability and Accountability Act; IEEE: Institute of Electrical and Electronics Engineers; NIST: National Institute of Standards and Technology; PCCP: Predetermined Change Control Plan; PHI: Protected Health Information; RMF: Risk Management Framework; SaMD: Software as a Medical Device; WHO: World Health Organization.

conference proceedings) and some preprints from arXiv.org to capture both established concepts and emerging techniques. While Scopus and Google Scholar provide broad interdisciplinary coverage essential for the intersection of AI engineering and health policy, we acknowledge that the exclusion of specialized biomedical databases (e.g., PubMed/MEDLINE) may limit the inclusion of some niche clinical validation studies. However, this strategy aligns with our primary focus on the technical and governance architectures of Trustworthy AI. The search which occurred between March 2025 to June 2025 used the following structured query:

*((“trustworthy” OR “trust” OR “transparent” OR “accountable” OR “responsible”) AND (“AI” OR “digital assistant” OR “LLM” OR “Large language Model” OR “artificial intelligence” OR “transformers” OR “Agentic”) AND (“healthcare” OR “health” OR “digital health” OR “medical” OR “medicine” OR “clinic”)).*

Initially, 450 search results were returned and only 70 of the research papers were eligible, as summarized in 2. The selection was restricted primarily to English-language publications and followed a two-stage screening process. The inclusion criteria encompassed studies that explicitly examined one or more of the considered dimensions of trustworthy AI in healthcare or related domains. Where

TABLE 2. Study selection process.

| Stage          | Description                                                                                                                           | Count |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------|-------|
| Identification | Initial search conducted on Scopus and Google Scholar (2017–2025).                                                                    | 450   |
| Screening      | Title and abstract review for relevance to Trustworthy AI and Healthcare. Removal of duplicates and non-English publications.         | ~380  |
| Eligibility    | Full-text assessment against inclusion criteria (explicit focus on Bias, Privacy, Robustness, or Explainability in medical contexts). | 70    |
| Inclusion      | Total initial selection of papers included in the integrative review before snowballing.                                              | 70    |

relevant, non-healthcare studies were also included if they introduced transferable frameworks or mitigation techniques (e.g., fairness optimization, privacy preservation) that were missing or insufficiently represented in healthcare contexts. The exclusion criteria removed duplicates, non-English materials, non-academic sources, and papers lacking substantial methodological or empirical contributions. The first stage of the process involved title and abstract review to determine topical relevance, followed by the second stage of full-text analysis to assess methodological rigor and applicability. Finally, reference chaining (snowballing) was performed to capture additional highly relevant studies cited by the included works. This method extended the materials used to much earlier years.

## G. SURVEY POSITIONING AND UNIQUE CONTRIBUTIONS

Previous trustworthiness literature, as summarized in Table 3, falls largely into two categories: General TAI Reviews ([26], [47]) that provide broad, foundational coverage of all key dimensions (Bias & Fairness, Privacy & Security, Robustness & Safety, and EIT) across multiple domains, and Healthcare-Focused TAI Reviews ([48], [49], [50]) which specialize in medical applications and prioritize concerns like safety and data confidentiality in traditional ML/DL systems. However, there exists a significant gap at the intersection of healthcare and emerging, complex AI paradigms like LLM and Agentic AI. Only a few isolated works, such as [51] on LLMs, indicate a shift toward these advanced models, but none provide a unified, comprehensive synthesis of TAI challenges within the healthcare domain for both LLM and Agentic AI.

To the best of our knowledge, this survey offers one of the first comprehensive analyses and classification of trustworthiness challenges and solutions specifically tailored to the application of both LLMs and Agentic AI in healthcare. By focusing on this high-stakes domain and these next-generation AI architectures, our work moves beyond the established concerns of general ML/DL to address novel risks such as model hallucination, autonomous decision-making in clinical workflows, and the unprecedented data requirements of foundation models. This positioning allows us to provide a timely and forward-looking roadmap for researchers and practitioners building the future of trustworthy medical AI.

We organize the rest of the survey as follows: Section II introduces the foundational dimensions of TAI, including bias and fairness, privacy and security, robustness and safety, and explainability, interpretability, and transparency. This section also examines trustworthiness in GenAI models, specifically LLMs, as well as in agentic AI. It also explores the intersectionality and trade-offs among these various dimensions. Section III surveys existing methods and strategies used to operationalize TAI across privacy, explainability and transparency, robustness, and fairness. In Section IV, we discuss the ongoing challenges of TAI, identify current research gaps, and discuss future directions for the field. Finally, Section V concludes the report by summarizing key issues and offering recommendations for advancing trustworthy AI in healthcare and beyond.

## II. TRUSTWORTHINESS CHALLENGES IN HEALTHCARE

Drawing from the ethical guidelines of the European Union for AI, a trustworthy AI system is fundamentally underpinned by four core ethical principles: respect for human autonomy, prevention of harm, fairness, and explicability [63]. These dimensions subsequently lead to seven more concrete requirements that AI systems must adhere to, encompassing human agency and oversight, technical robustness and safety, privacy and data governance, transparency, diversity, non-discrimination and fairness, societal and environmental well-being, and accountability. This multi-dimensional framework

recognizes that sole reliance on accuracy is insufficient for responsible and reliable AI development. This survey specifically delves into four key dimensions vital for achieving trustworthy AI in practical applications in healthcare: Bias & Fairness, Privacy & Security, Robustness & Safety, and Explainability, Interpretability & Transparency, offering a comprehensive assessment of associated trustworthy issues in different healthcare contexts.

## A. TRUSTWORTHINESS IN TRADITIONAL MACHINE LEARNING AND DEEP LEARNING

### 1) BIAS AND FAIRNESS

Bias is defined as systematic differences in the performance of models across different subgroups of the population. Similarly, fairness refers to the ability of AI models to identify, mitigate, and manage biases in both data and algorithms, ensuring that decisions and outcomes are made without favoritism or discrimination [1], [15], [64], [65], [66], [67].

As AI becomes integrated in sensitive domains such as healthcare, ensuring fairness and avoiding discriminatory behaviors in AI systems is critical to maintaining trust and adoption. According to [68], the performance and fairness of AI models in healthcare are heavily influenced by the quality and diversity of their training data. For instance, models trained on data from specific hospitals, regions, or devices may not generalize well to other settings with different conditions. Additionally, these models can pick up demographic cues such as race or gender from medical images, which can lead to biased decision-making or diagnostic shortcuts. Chest radiograph classifiers have been shown to exhibit biased performance, underdiagnosing conditions in populations such as black women, which risk delayed treatment and exacerbated health inequalities [69]. Furthermore, dermatology-focused AI models have been shown to have exhibited reduced accuracy in diagnosing conditions such as melanoma among darker-skinned individuals, a disparity directly attributable to their training on datasets predominantly composed of images from fair-skinned individuals [70], [71], [72]. In computer vision, facial recognition systems perform better on white males [73], and biased object detection models have also been reported [74]. Similar biases have been observed in NLP, and speech recognition systems [75], [76], [77]. In addition, LLMs have shown discrimination against females, individuals of the black race, and a disproportionately higher association of violence with Muslims [27], [78], [79]. [80] found that LLMs used in healthcare exhibit significant socio-demographic biases, disproportionately recommending more urgent, invasive, or mental-health interventions for marginalized groups such as Black, unhoused, and LGBTQIA+ patients even when clinical scenarios are identical.

Reference [1] classifies bias as productive (necessary for generalization), erroneous (from flawed assumptions or sampling), and discriminatory (causing unequal treatment).

TABLE 3. Positioning and scope of Trustworthy AI (TAI) literature comparison.

| Previous Work | Domain Focus                                                                                           | TAI in ML/DL Addressed? | TAI in LLMs Addressed? | TAI in Agentic AI Addressed? | Addressed Intersections? | Focused Trustworthiness Dimensions                                                                                                 |
|---------------|--------------------------------------------------------------------------------------------------------|-------------------------|------------------------|------------------------------|--------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| [26]          | Diverse (Transportation, Finance, Medicine, Security, Entertainment, Industrial applications)          | Yes                     | No                     | No                           | Yes                      | Robustness & Safety; Explainability, Interpretability & Transparency; Bias & Fairness; Privacy & Security; Accountability.         |
| [47]          | Diverse (Education, Environmental Science, 5G-IoT, Robotics for AEC, Financial Technology, Healthcare) | Yes                     | No                     | No                           | Yes                      | EIT; Bias & Fairness; Robustness & Safety; Privacy & Security; Accountability, Accuracy, Reproducibility, Human Agency, Oversight. |
| [52]          | Diverse (Banking, Healthcare, Autonomous Systems, IoT, diverse industries)                             | Yes                     | No                     | No                           | Yes                      | EIT (Primary focus); Robustness & Safety; Privacy & Security; Bias.                                                                |
| [51]          | Diverse (Finance, Healthcare, Education, Policy, Legal Systems)                                        | Yes                     | Yes                    | No                           | Yes                      | EIT; Robustness & Safety; Privacy & Security; Bias & Fairness.                                                                     |
| [48]          | Healthcare/Medical                                                                                     | Yes                     | No                     | No                           | Yes                      | EIT; Robustness & Safety; Privacy & Security; Bias & Fairness.                                                                     |
| [53]          | Healthcare systems                                                                                     | Yes                     | No                     | No                           | Yes                      | EIT (Primary focus); Robustness & Safety; Privacy & Security; Bias & Fairness.                                                     |
| [49]          | Healthcare                                                                                             | Yes                     | No                     | No                           | Yes                      | EIT (focus area); Robustness & Safety; Privacy & Security; Bias & Fairness; Accountability.                                        |
| [54]          | Healthcare (Medical AI)                                                                                | No                      | No                     | No                           | Yes                      | Bias & Fairness; Privacy & Security; EIT; Data quality, Responsibility attribution.                                                |
| [50]          | Healthcare/Medicine                                                                                    | Yes                     | No                     | No                           | Yes                      | EIT (Primary focus); Bias & Fairness; Privacy & Security.                                                                          |
| [55]          | Healthcare                                                                                             | Yes                     | Yes                    | No                           | Yes                      | Bias & Fairness; Privacy & Security; EIT; Robustness & Safety.                                                                     |
| [56]          | Healthcare (Cardiovascular care)                                                                       | Yes                     | No                     | No                           | Yes                      | Privacy & Security; Bias & Fairness; Robustness & Safety; EIT; Accountability/Responsibility.                                      |
| [57]          | Diverse (Healthcare, Finance, Transportation, Entertainment)                                           | Yes                     | No                     | No                           | No                       | EIT; Bias & Fairness; Privacy & Security; Accountability/Responsibility.                                                           |
| [58]          | Healthcare                                                                                             | Yes                     | No                     | No                           | No                       | Robustness & Safety; EIT; Bias & Fairness.                                                                                         |
| [59]          | Healthcare                                                                                             | Yes                     | No                     | No                           | No                       | Bias & Fairness; EIT.                                                                                                              |
| [60]          | Healthcare                                                                                             | No                      | Yes                    | Yes                          | Yes                      | Truthfulness; Privacy & Security; Robustness & Safety; Fairness & Bias; EIT.                                                       |
| [61]          | Healthcare                                                                                             | Yes                     | Yes                    | Yes                          | Yes                      | Fairness & Bias; Accountability; Privacy & Security; Robustness & Safety; Interpretability; Explainability & Safety.               |
| [62]          | Healthcare                                                                                             | No                      | Yes                    | No                           | Yes                      | Biases, Reliability, Safety, EIT, Robustness.                                                                                      |
| This survey   | Healthcare                                                                                             | Yes                     | Yes                    | Yes                          | Yes                      | <b>Bias &amp; Fairness; Privacy &amp; Security; EIT; Robustness &amp; Safety.</b>                                                  |

Note: 5G: Fifth Generation Technology; AEC: Architecture, Engineering, and Construction; DL: Deep Learning; EIT: Explainability, Interpretability, and Transparency; IoT: Internet of Things; LLM: Large Language Model; ML: Machine Learning; TAI: Trustworthy AI.

Bias can originate from data annotation (e.g., biased, non-diverse or inexperienced annotators), collection (e.g., skewed sources), or processing stages. Reference [15] further categorizes it as data bias (unrepresentative), algorithm bias (flawed learning processes), explicit bias (systemic inequities), implicit bias (unconscious patterns), and selection bias (not generalized). These biases can lead to loss of trust in healthcare systems, particularly when AI-driven decisions appear opaque or unfair, discouraging patients, especially in marginalized communities from engaging in AI-based care, which results in what is referred to as *under-trust* [34]. Table 10 summarizes these bias sources across the pipeline (Data, Algorithmic, Evaluation), serving

as a structured taxonomy for practitioners. Additionally, legal and ethical implications arise when biased AI violates anti-discrimination laws or breaches medical ethical principles such as justice and non-maleficence, potentially exposing healthcare providers to litigation and eroding patient autonomy.

Bias mitigation strategies fall into three categories: pre-processing (e.g., resampling, reweighting, blinding, relabeling, adversarial data generation); in-processing (e.g., regularization, adversarial debiasing learning, fairness-aware learning, constraint-based optimization, and multi-armed bandits); and post-processing (e.g., threshold adjustment, output recalibration). Each approach to addressing fairness

in algorithms involves inherent trade-offs among fairness, accuracy, and complexity. In-processing methods, for example, aim to strike a balance between predictive performance and fairness but may require complex model modifications. Post-processing techniques highlight the tension between maintaining prediction accuracy and ensuring calibration, as adjustments after training can inadvertently compromise one objective for the other. More broadly, efforts to enhance fairness often come at the expense of model performance, since improvements in equity may reduce predictive accuracy or efficiency [1], [20].

In the context of Conversational AI (CAI), bias mitigation must extend beyond algorithmic outputs to address cognitive and behavioral factors influencing users and stakeholders. As [35] noted, stakeholders across the CAI ecosystem, including users, designers, and developers, are susceptible to cognitive biases, heuristics, and information asymmetry, which can compromise privacy and security. To counter these effects, the authors propose a multidimensional debiasing framework that includes educational initiatives, cognitive nudges, decision-support tools, incentive structures, and optimized default settings.

Meanwhile, fairness is categorized into group fairness (ensuring statistical parity between groups) and individual fairness (ensuring similar individuals receive similar outcomes) [1]. According to [30], the goal is to ensure that a model treats different groups, defined by an attribute such as age or gender, equitably by equalizing its prediction behavior across those groups. There are two primary perspectives on fairness: procedural fairness emphasizes equal treatment regardless of group identity, whereas substantive fairness allows differential treatment to achieve equitable results. These approaches aim to address disparate treatment and impact. Procedural fairness may inadvertently lead to disparate treatment, whereas substantive fairness addresses disparate impact, where certain groups receive less favorable outcomes despite equal treatment.

Fairness evaluation is challenging due to context dependence and varying stakeholder subjectivity. To address this, [33] presents various evaluation strategies and metrics, including data analysis, which examines class distributions and subgroup partitions, visualizes attribute distributions, and inspects data samples to detect quality issues or biases. Group fairness evaluation assesses whether groups defined by sensitive variables (e.g., gender or ethnicity) are treated equitably. Some of the common metrics of measuring fairness include Demographic Parity (equal positive classification probability across groups), Equal Opportunity Difference (EOp - equal true positive rates (TPR)), and Equalized Odds Difference (EOd - equal TPR and false positive rates across groups). EOd ensures equitable likelihoods of correct identification mitigates the disproportionate impact of false alarms across all patient cohorts [66].

In clinical decision-support settings characterized by imbalanced labels and continuous risk scoring, metric families such as Area Under the Receiver Operating

Characteristic curve (AUROC), Area Under the Precision-Recall Curve (AUPRC), Brier scores, calibration curves, and equalized-odds-based fairness measures are more appropriate than accuracy-based metrics. These families better capture nuanced risk stratification and support equitable decision thresholds across heterogeneous patient populations.

Other measures like Disparate Impact, Predictive Parity, and Test Fairness offer additional perspectives by comparing ratios, accuracy, or probability distributions between subgroups. Individual fairness evaluation, though less frequently applied, focuses on ensuring that similar individuals receive similar outcomes. It includes concepts like fairness through unawareness, fairness through awareness, the Generalized Entropy Index (variation in individual prediction accuracy), and Counterfactual Fairness (consistency of decisions if sensitive attributes were changed).

Reference [30] quantified fairness through accuracy disparity, calculated as follows:

$$\Delta_{\text{acc}} = \max_{a,b \in A} |\text{acc}(F, D_a) - \text{acc}(F, D_b)| \quad (1)$$

where  $D_a$  and  $D_b$  are subsets of the dataset belonging to groups  $a$  and  $b$ , and  $\text{acc}(F, D)$  is the accuracy of model  $F$  on dataset  $D$ . Minimizing  $\Delta_{\text{acc}}$  is a typical fairness objective and can be pursued through various interventions as outlined above (pre-processing, in-processing, and post-processing methods).

Different governance frameworks operationalize fairness and safety in distinct ways. For instance, HLEG emphasizes demographic and procedural fairness, the U.S. AI Bill of Rights prioritizes disparate-impact avoidance, and NIST focuses on measurement and risk characterization. In clinical datasets, these translate into varied fairness targets, such as demographic parity, equal opportunity, or predictive parity, and different safety validation protocols ranging from Out-of-Distribution (OOD) stress testing to formal hazard analysis.

## 2) PRIVACY AND SECURITY

Privacy in AI systems refers to the ability of AI systems to safeguard sensitive personal information across all stages, including training, deployment, and inference, ensuring the model remains protected against data leakages or unauthorized disclosures. Legal frameworks like GDPR provide structured guidance on data collection, processing, and usage. In practice, data governance ensures accountability and defines rules for handling data and algorithms responsibly. Evaluating privacy and data governance is vital to prevent breaches, misuse, and re-identification risks, and to sustain user trust in AI [33], since AI models can unintentionally memorize and leak sensitive information from their training data [81], [82]. Chat systems have been observed, for instance, to recall personal identifiers and medical details, especially if these appeared frequently during training [1].

### a: VULNERABILITY AGAINST PRIVACY ATTACKS

Various attacks, such as membership inference, attribute inference, model inversion, and model extraction, target different privacy facets [36]. Membership inference for instance, determines whether a data point was part of the training set, often exploiting overfitting [1], [13], [30], [83], [84]. While attribute inference deduces hidden features based on learned correlations, model inversion reconstructs individual data from outputs, and model extraction replicates proprietary models to steal intellectual property. These attacks are evaluated by the performance of attack models, including accuracy, F1-score, Attack Success Rate, and fidelity metrics, among others [33].

In healthcare, these vulnerabilities manifest through specific vectors, such as the leakage of patient information via Digital Imaging and Communications in Medicine (DICOM) header leakage [85], adversarial manipulation of medical images [86], poisoning of modality-specific pipelines [87], and genomic re-identification attacks [88]. Privacy and security concerns have been analyzed in Section III-A and summarized in Table 6.

### 3) ROBUSTNESS AND SAFETY

Robustness and safety are critical for TAI, particularly in healthcare where model failures can lead to severe consequences like misdiagnosis or inappropriate treatments [1], [29], [33]. Consider unpredictable medical scenarios, where patient conditions can suddenly change or new diseases may emerge; the robustness of an AI system is crucial to ensure accurate and trustworthy insights. For instance, perturbations like slight pixel-level changes in a chest X-ray or MRI scan could cause a model to misclassify a tumor as benign instead of malignant, and manipulation of values in a patient's EHR (e.g., slightly adjusting lab test results) could prevent an AI model from detecting signs of sepsis or cardiovascular risks.

Robustness to adversarial attacks evaluates a model's resilience to malicious perturbations that alter inputs subtly but significantly. These attacks are often categorized based on attacker goals (targeted vs. untargeted), knowledge (white-box vs. closed box), and methods (evasion vs. poisoning) [1], [33]. For instance, poisoning attacks occur during training, where attackers insert malicious samples to corrupt the model's behavior or introduce backdoors. Evasion attacks, in contrast, occur during inference by subtly modifying test inputs to mislead the model. Glass box attacks assume full access to the model's internals, whereas closed box attacks rely only on model outputs. While targeted attacks aim for specific incorrect outputs, non-targeted attacks only require that the output be wrong. Various architectures remain vulnerable to adversarial attacks such as poisoning or evasion. For instance, CNN can be misled by pixel-level changes, Graph Neural Networks (GNN) can be tricked by altering node features or graph structure, and RNN models in NLP can be manipulated through subtle linguistic modifications. These vulnerabilities span multiple domains, showing that no model architecture is inherently robust [1].

Reference [33] classifies robustness into distribution shift and adversarial robustness, extending analysis to uncertainty estimation and OOD detection. Distribution shift types include covariate shift, also known as data shift (changes in input features with constant target label distribution), label shift (changes in target label distribution), or concept drift (combined shifts in features and labels over time). For instance, a model trained to predict heart attack risk on data from an older, sicker population and then used on a younger, healthier population is an example of covariate shift, as the distribution of patient features changes whereas the relationship between those features and heart attack risk remains the same. Meanwhile, the COVID-19 pandemic is an example of concept drift because it fundamentally changed the relationship between a person's symptoms and the underlying disease, making pre-existing models for respiratory illnesses obsolete.

These are tested via synthetic corruption, or varying demographics, or sensor variance, using metrics like accuracy drop or regret [13], [26], [89]. OOD detection focuses on recognizing inputs that fall outside the scope of what the model was trained on, helping to prevent incorrect predictions on unfamiliar data. Methods such as maximum softmax probability, the ODIN technique, energy and confidence scores (e.g., Mahalanobis distance) are used for this purpose. Some of the challenges with this technique include overconfident predictions and discrepancies between test datasets and real-world OOD data, which can hinder detection performance. To address these issues, uncertainty evaluation, typically categorized into aleatoric (data-driven) and epistemic (model-related) types, provides a framework for assessing how well an AI system can quantify and express confidence in its predictions. Some of the metrics include Negative Log-likelihood (Log Loss), Brier Score, Expected Calibration Error (ECE), Maximum Calibration Error (MCE), Area Under the Sparsification Error Curve (AUSE), and Mutual Information (MI). A list of key metrics used to quantify and express the aleatoric and epistemic uncertainty in AI predictions is provided in Table 10.

### 4) EXPLAINABILITY, INTERPRETABILITY AND TRANSPARENCY

Explainability, interpretability, and transparency are closely related yet distinct concepts crucial for building trust in AI systems, particularly in healthcare. They are often collectively operationalized through XAI techniques. Explainability refers to the process of rendering the decision-making of a model understandable to humans, especially in closed box systems like deep neural networks. It can also be described as the process of providing a justification for the output of a model. Interpretability is the understanding of the output and internal workings of a model, in human terms with respect to its inputs [90], [91]. Meanwhile, transparency includes explainability and interpretability, which describes how openly model-related information, such as features, performance, limitations, data sources, labeling protocols,

and assumptions, are shared with stakeholders [21], [33]. While explainability often involves convincing the user of the validity of model prediction, interpretability is more focused on the exploration of model properties to provide transparency [90], and a model is deemed interpretable if humans can directly understand its predictions [1].

Reference [90] classify interpretability evaluation into three types: application-grounded (real-world tasks with experts), human-grounded (simplified tasks with general users), and functionally grounded (mathematical metrics without human input), providing a foundation for applying explanation methods across contexts. Not all explanation modalities are equally feasible in clinical workflows. Functionally grounded metrics enable automated validation but offer limited clinician utility; human-grounded evaluations better reflect end-user comprehension but introduce scalability challenges due to the complexity of expert decision-making; and application-grounded evaluations, though most clinically authentic, require significant expert time [92], [93], [94].

XAI supports model validation, debugging, and bias detection. It identifies when models rely on irrelevant inputs or biased training data. According to [1] and [20], explainability is essential for trust, especially in high-stakes domains like healthcare, by clarifying how decisions are made, what data was used, and how outputs are validated.

Explainability finds diverse applications in healthcare, including drug discovery and disease diagnosis [1]. Graph Neural Networks (GNN) have identified mutagenic groups in molecules relevant to bacteria like *Salmonella typhimurium*. In clinical decision-making, AI explanations must reflect the context of the patient and complexity of the disease. Some libraries like AIX360, InterpretML, DeepExplain, and DIG support XAI across domains [1], [26], [33]. However, a persistent challenge is ensuring that explanations are comprehensible to non-technical users, such as clinicians.

The evaluation of explainability is complex due to the varied needs of the stakeholders. Domain experts may require attribute-level logic, policy makers need system-level insights, and lay users often seek concise justifications. To address this, explanations are categorized by audience, content, and generation method. The ISO provides criteria for evaluating the quality of the explanation, including consistency (importance should match behavior), continuity (similar inputs yield similar explanations), and selectivity (focus on most influential features). Additional frameworks, such as fact sheets, assess explanation methods across dimensions like usability, safety, and validation [20].

## B. TRUSTWORTHINESS IN GENERATIVE AI MODELS

Generative AI is emerging as a transformative force in AI, with the capability of producing multimodal content including text, images, videos, music, and software code; based solely on prompts. It is reshaping the healthcare landscape by enabling the creation of diverse content such

**TABLE 4. LLM integration lifecycle in healthcare.**

| Phase                      | Key Tasks & Considerations                                                                                                                                                   |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Planning                   | Scope; clinical alignment; privacy governance; regulatory review; explainability; IT infra assessment; early clinician involvement [25], [95]                                |
| Data Collection & Curation | Representativeness; data quality; breadth; workflow alignment; privacy; governance; high-quality, representative datasets [15], [30], [95]                                   |
| Model Dev. & Validation    | Patient-centric; clinician-in-loop; performance; interpretability; scalability; uncertainty; hallucination control; compliance; use RLHF, fine-tuning [21], [25], [82], [95] |
| Deployment                 | Outcomes monitoring; user feedback; CDS impact; privacy; interpretability; workflow impact; pilot in controlled settings; iterate for reliability [25], [26], [95], [96]     |
| Monitoring & Maintenance   | Ongoing clinician collaboration; health outcome tracking; workflow monitoring; feedback; performance; scalable infra; continuous refinement [21], [25], [95]                 |

*Note:* CDS: Clinical Decision Support; IT: Information Technology; RLHF: Reinforcement Learning from Human Feedback.

as clinical notes, diagnostic reports, medical images, patient summaries, and even predictive models. Unlike traditional AI that has focused primarily on decision-making (diagnostics) and prescriptive analytics, GenAI aids in the creation and interpretation of medical content, as seen in the use of large language models like GPT to generate patient histories, suggest treatment plans, or draft discharge summaries based on EHRs [67].

## 1) OVERVIEW OF LARGE LANGUAGE MODELS

### a: LANGUAGE MODELS (LM)

These are models that are trained to understand and generate human language by modeling the statistical structure of word sequences. Fundamentally, they predict the likelihood of a token sequence using the chain rule of probability. They typically use autoregressive methods to estimate the conditional probability of each token based on prior tokens. Transformers, especially encoder-only, decoder-only, and encoder-decoder architectures, are the standard backbone of these models, widely applied in tasks such as text generation, information retrieval, and question answering [27].

### b: LARGE LANGUAGE MODELS

LLMs are an advanced extension of pre-trained language models (PLMs) characterized by their massive scale in terms of billions of parameters trained on vast text corpora, and computational resources. Most modern LLMs use decoder-only transformer architectures and demonstrate emergent abilities like in-context learning (ICL), chain-of-thought reasoning, and instruction following. Some examples include the GPT series (like ChatGPT), Gemini, LLaMA, Claude and Mistral, to mention a few. Meanwhile, in transformers, the key innovation is the self-attention mechanism, which enables models to process sequences in parallel and capture long-range dependencies more effectively than earlier architectures. Table 4 highlights some key considerations

**TABLE 5. Strategies for adapting general LLMs to healthcare.**

| Category                         | Approaches                                                                                                    |
|----------------------------------|---------------------------------------------------------------------------------------------------------------|
| Pre-trained Model Foundation     | Choose base models by size, corpus relevance, prior healthcare performance [65], [95], [97]                   |
| Domain Adaptation Techniques     | Prompt engineering; chain-of-thought; fine-tuning (LoRA) or QLoRA; RLHF [26], [29], [95], [98]                |
| Knowledge Transfer & Enhancement | LLM embeddings; structured data integration; cross-domain knowledge preservation [32], [95], [97]             |
| Data-Centric Enhancements        | Medical data augmentation; rare-case simulation; external knowledge base (KB) integration [24], [32], [95]    |
| Multi-Agent LLMs                 | Role-specific agents; workflow simulation; multi-step decision support [31], [32], [99]                       |
| Robustness & Efficiency          | Adversarial training; uncertainty quantification; privacy (DP, anonymization); distillation [36], [99], [100] |
| Continuous Adaptation            | Knowledge updates; feedback-driven iteration; performance monitoring; versioning [25], [82], [95]             |

*Note:* DP: Differential Privacy; KB: Knowledge Base; LoRA: Low-Rank Adaptation; QLoRA: Quantized Low-Rank Adaptation; RLHF: Reinforcement Learning from Human Feedback.

for different stages of LLM lifecycle in healthcare, whereas Table 5 summarizes some of the strategies for integrating LLMs in healthcare, as suggested by [95].

### c: MULTIMODAL LARGE LANGUAGE MODELS (MLLM)

MLLM on the hand, generates multiple data types, like text, images, audio and videos. They enhance LLMs by integrating visual inputs through visual encoders and vision-to-language adapters. These models are capable of handling complex multimodal tasks like image captioning, visual dialogue, and visual question answering by combining textual and visual data streams. Other Foundation Models, such as Diffusion Models (DMs), are prominent in the domains of image and video generation. These models work by learning to reverse the process of noise addition to generate data, as seen in models like DALL-E, Stable Diffusion (for images), and Sora (for videos) [27].

### 2) ALIGNMENT

Alignment is the process of ensuring that LLMs behave consistently with human values, ethical standards, and societal norms. This includes guiding outputs toward being fair, safe, honest, and useful, particularly in high-stakes settings like healthcare. Early models like GPT-3 demonstrated impressive capabilities, but also exhibited harmful behaviors, such as hallucinations, biased outputs, and unsafe content, due to their exposure to massive, unfiltered internet data. As a result, alignment has become central to improving trust and adoption [101], [102], [103], [104]. Reinforcement Learning from Human Feedback (RLHF) is a core method for aligning LLMs. It involves human feedback to rank responses, which trains a reward model used to guide further tuning. Supervised Fine-Tuning (SFT), often applied before RLHF, uses curated examples to instill desired model behaviors.

More recent strategies incorporate hybrid feedback (human + AI), but all aim to reduce toxicity and bias while improving reliability [27], [65]. The widely referenced HHH principle - Helpful, Honest, and Harmless, encapsulates alignment goals, although operationalizing it remains challenging [102]. Alignment also intersects with core trust dimensions such as reliability, safety, fairness, explainability, and robustness.

### 3) RELIABILITY AND HALLUCINATION

Reliable LLMs are expected to produce factually accurate, stable, and coherent outputs. However, challenges like misinformation and hallucination remain prevalent, as LLMs can generate content that sounds plausible but lacks factual grounding, often due to poor quality training data or the over-reliance on memorized patterns, contradicting the original input or real-world information [23], [65].

The integration of AI into healthcare is critically threatened by clinical hallucination risk, where models generate plausible but factually incorrect medical content. Given the minimal margin of error in medicine, a single error like a fabricated drug interaction can lead to life-threatening outcomes, with experts classifying many such inaccuracies as posing significant or considerable clinical risk, potentially scaling to catastrophic patient harm [105]. This danger is crystallized in the problem of on-label vs. off-label use. For instance, a factual hallucination might falsely claim a drug is FDA-approved for a specific condition, directly prompting off-label or inappropriate use. This risk is amplified by incomplete Intended Use Statements (IUSs) that fail to specify limitations, leading to inadvertent use on unvalidated populations, like children, and risking patient injury or death [106].

Inconsistency also arises when models produce different answers to the same question due to random sampling or session differences. Another challenge is miscalibration, where LLMs express unwarranted confidence in uncertain or outdated topics. LLMs sometimes exhibit sycophantic behavior by overly agreeing with user input, especially when pressured. This stems from instruction tuning that prioritizes helpfulness over correctness, risking reinforcement of user misconceptions.

Beyond descriptive accounts, hallucination risk can be examined through quantitative models of misinformation generation. Recent modeling approaches have begun to quantify how misinformation mechanisms affect systemic trust. For instance, generative AI models have been shown to mathematically capture how user awareness, emotional contagion, and belief alignment amplify the propagation of misinformation, linking generative errors directly to systemic trust decay [107].

### 4) PRIVACY AND SAFETY

Despite alignment efforts, LLMs can still generate or reveal unsafe, harmful, or inappropriate content, ranging from violence and explicit material to illegal advice. They can also

memorize sensitive information from training data, posing privacy risks [81], [82]. Attacks like membership inference or model inversion can exploit these vulnerabilities. While privacy-enhancing technologies such as differential privacy and federated learning (FL) offer partial defenses, their scalability to large-scale LLMs remains limited. To promote privacy and safety, [65] propose guardrails to dynamically guide LLM input and output. This is further corroborated by [27] that introduced a modular guardrail system which wraps safety-focused input and output filters around LLMs to intercept harmful content, offering flexible and decoupled protection despite limitations against advanced attacks and added computational overhead.

In an effort to address privacy-utility trade-off in LLMs, [83] provides a comprehensive Systematization of Knowledge (SoK) on privacy risks and mitigation strategies in LLMs, focusing especially on challenges arising from user interactions and advanced LLM capabilities. It categorizes privacy risks into four domains: training data (e.g., data memorization and extraction attacks), prompts (e.g., user input leakage and inference), outputs (e.g., disclosure of sensitive info), and LLM agents (e.g., autonomous behaviors exposing data). It outlines a range of mitigation techniques, including differential privacy, prompt sanitization, encrypted inputs, output obfuscation, and sandboxed agent environments.

In LLM-based chatbots, for instance, [98] proposes and compares two architectures for a privacy-preserving chatbot for hypertensive patient self-management. The first is a hybrid approach combining a classifier to detect sensitive inputs with GPT-3.5 Turbo for non-sensitive queries, offering strong performance but risking privacy breaches if misclassification occurs. The second uses fully local open-source LLMs (e.g., Llama2, Mixtral) to ensure complete data privacy, although it requires significant resources and domain-specific fine-tuning. In general, the study finds open-source LLMs to be a promising direction for secure, effective medical chatbots.

## 5) FAIRNESS AND BIAS

Similar to traditional machine learning systems, LLMs may replicate or even amplify societal biases, affecting users across gender, race, or culture. However, one difference is that LLMs are pre-trained on massive, uncensored internet data, which has embedded societal biases. The key forms of bias include stereotyping based on demographic attributes, disparate performance across languages or regions, and preference bias, where ideological leanings influence output. For example, an LLM-powered clinical support system may recommend different treatment pathways for male versus female patients presenting the same symptoms (e.g., chest pain). Historically, women's cardiovascular symptoms have been underrepresented or mischaracterized in medical literature. If the model inherits this bias, it could underestimate the likelihood of heart disease in women, leading to delayed diagnosis or inappropriate care.

Similarly, an LLM integrated into a telehealth chatbot may perform less accurately in Spanish, Mandarin or other related languages compared to English, since English dominates the training data. This can lead to misinterpretation of patient-reported symptoms, such as describing pain or mental health concerns, ultimately worsening disparities in care for non-English-speaking patients. Therefore, some mitigation approaches include realignment to make models helpful, honest, and harmless [102]. Other methods include data balancing, bias-aware training, and fairness-aware evaluation techniques.

## 6) EXPLAINABILITY

Trustworthy LLMs must offer insights into their decision-making process. Explainability in traditional ML relies on interpretable structures and straightforward post-hoc techniques, whereas explainability in LLMs involves complex, often opaque mechanisms that involve advanced prompting, interpretation of internal representations, among others. Newer models like ChatGPT and Gemini enable more interactive and dialog-based explanation formats. Techniques like Chain-of-Thought (CoT) prompting and Retrieval-Augmented Generation (RAG) aim to make reasoning more transparent, though they often rely on pattern recognition rather than true logical or causal inference [101], [102], [103].

LLMs become increasingly embedded in real-world applications, XAI evolves from merely unraveling the closed box nature of models to forming a synergistic relationship with LLMs. XAI improves the productivity and applicability of LLMs by providing interpretability, whereas LLMs in turn, through their conversational and generative capabilities, make XAI more user-friendly and intuitive. This mutual enhancement allows explanations to be more accessible and actionable, integrating common sense and domain knowledge to boost both development and deployment [32]. One core approach to enhancing LLMs is through attribution methods, which diagnose and improve models by identifying how input features influence predictions. These post-hoc techniques quantify the importance of input words in tasks such as text classification or text generation. Classification-focused methods include perturbation-based explanations, which alter input tokens to observe output changes, though they are computationally intensive. Gradient-based methods offer a more efficient alternative by measuring sensitivity through backpropagation. Surrogate-based explanations approximate decision boundaries with simpler models, whereas decomposition-based techniques assign additive relevance scores, each carrying trade-offs in scalability and generality [32], [108].

In generative settings, interpretation must consider the sequential nature of output. Word-level classification analogies are applied, with techniques like Mask Perturbation and Integrated Gradients evaluating token influence. Some strategies capitalize on LLMs' own generative abilities to cite supporting passages, either from external sources or their internal

memory, though these raise concerns around hallucination and necessitate benchmarks for faithfulness. Attributions not only explain outputs but also improve usability by enabling model diagnosis and prompt refinement [32]. For example, frameworks like RARR [109] detect unsupported claims, and methods such as AMPLIFY [110] use attribution to enhance future prompt engineering. Nonetheless, attribution-based methods face ongoing challenges in efficiency and faithfulness, especially for generative LLMs.

Beyond surface-level attribution, interpreting model internals offers deeper insight into LLM behavior. By analyzing the self-attention and feed-forward components of the transformer, researchers can trace how layers influence outputs. Techniques like residual stream decomposition, component importance analysis, and circuit analysis reveal how the model forms predictions, aiding tasks such as knowledge editing and model pruning to enhance efficiency and interpretability [32], [108].

Explainable prompting improves inference transparency by guiding models through step-by-step reasoning, as seen in Chain-of-Thought (CoT) prompting. Extensions like Tree-of-Thoughts (ToT) and Graph of Thoughts (GoT) enable more complex reasoning. Studies show that CoT's faithfulness varies by model, highlighting the need to align explanations with actual reasoning and avoid over-reliance on templates. Knowledge-augmented prompting, especially through RAG, enhances interpretability by grounding outputs in retrieved documents. This reduces hallucinations, supports timely updates, and enables domain-specific responses. While RAG is valuable for reducing hallucinations and incorporating current information, it also introduces new risks, including robustness issues, privacy concerns, adversarial attacks, and accountability problems [84].

Furthermore, discussion on interpretability must differentiate between the needs of developers versus clinical end-users. While developers require attribution methods for model debugging, clinicians require verification of faithfulness, ensuring that the explanation (e.g., CoT reasoning) accurately reflects the model's actual decision process rather than just providing a plausible post-hoc justification [111], [112], [113], [114].

## 7) ROBUSTNESS

LLMs are vulnerable to input manipulations, such as prompt attacks that subtly change model output [102]. They are also susceptible to distribution shifts, making outputs outdated or contextually incorrect, and adversarial techniques like jailbreaking and prompt injection, which bypass safeguards or leak system instructions. Poisoning attacks, where adversaries corrupt training data with malicious intent, also pose significant risks. To harden models against attacks, some techniques have been proposed, including adversarial training, data sanitization, and red-teaming. However, robustness against unknown or evolving threats remains an open challenge [95].

## C. TRUSTWORTHINESS IN AGENTIC AI

Agentic AI refers to autonomous systems designed to perform complex, long-term tasks without human supervision by learning from context and making decisions independently. Unlike classical AI which relies on fixed rules or supervised learning for specific, narrowly defined tasks, Agentic AI systems exhibit adaptability and autonomy, enabling broader and more dynamic functionality [99]. In healthcare, for example, a classical AI system might identify pneumonia in chest X-rays based on predefined patterns and labeled training data, but it would struggle with rare presentations or evolving disease patterns. In contrast, an Agentic AI system could continuously integrate new patient data, adapt to emerging conditions, adjust its diagnostic reasoning in real time, and even plan follow-up tests or treatments, thereby mirroring the adaptive decision-making of human clinicians in complex, real-world scenarios.

Agentic AI differs significantly from traditional AI in both function and autonomy. While traditional AI systems are built for narrow, rule-based tasks and require significant human intervention, Agentic AI is designed for broader goals, operating with minimal oversight. It is highly adaptable, context-aware, and learns through reinforcement or self-supervision, enabling it to make complex decisions dynamically. In contrast, traditional models interact with static environments and follow predefined logic, limiting their responsiveness and scalability [31], [99].

Since agents often need continuous access to sensitive data and external systems to act effectively, they enlarge the attack surface and makes them vulnerable to privacy breaches and cyberattacks. They may inadvertently leak private data through interactions or be manipulated to extract information. For instance, a patient-care agent with access to medical records could be tricked by an adversarial prompt into revealing sensitive health details or could accidentally share identifying data while coordinating with third-party systems. To mitigate this, they require strict security standards, encryption and privacy-preserving techniques like differential privacy and homomorphic encryption. Access control, adversarial defenses, runtime monitoring and data anonymization can provide a multi-layered defense [31], [33], [97].

Furthermore, agentic AI can inherit and amplify biases from its training data, leading to unfair outcomes. Unlike static models, agentic AI does not just generate outputs, it can act on them in the real world. This means any bias in training data or reasoning gets translated into decisions and actions at scale. For instance, a biased diagnostic agent may not only misclassify certain populations (e.g., underdiagnosing heart disease in women) but also proactively schedule different follow-up care for those population subgroups, compounding disparities.

Similarly, as agentic AI makes autonomous, sequential decisions (not just single predictions), the reasoning process becomes even harder to trace or explain. Decision chains can grow opaque, especially if the agent adapts dynamically

based on feedback. In healthcare triage for example, an agent might escalate some cases and downgrade others based on hidden intermediate rules, leaving clinicians unable to reconstruct why the system acted differently across patients with similar profiles.

The high risk of clinical hallucination, off-label use and the potential for catastrophic patient harm create a crisis in responsibility attribution, with unclear chain of accountability which raises questions over whether liability falls on the AI developer, the healthcare provider, or the institution. To manage these interconnected risks, robust evaluation protocols acceptable to clinicians and regulators are imperative. Regulators demand compliance through transparent IUSs and advocate for new safeguards like model cards safety indicator, while clinicians must conduct iterative testing using standardized scenarios to verify outputs against the accepted standard of care and implement proactive error reporting to identify hallucinations before they cause harm [106], [115], [116].

Despite the perceived explainability benefits of CoT in LLM, its applications in agentic AI pipelines has been shown not to inherently lead to better output quality or improved explainability for end-users [117]. Manuvinakurike et al. demonstrates through a quantitative and qualitative analysis that CoT can produce what they called *explanations without explainability*, which fails to help users understand the system or achieve their goals, thereby adversely affecting user trust. Furthermore, the research finds that models trained without reasoning capabilities can actually perform better.

In terms of robustness and safety, agentic AI can magnify safety risks due to its ability to act with minimal human oversight. While a traditional AI might give a flawed recommendation, a human is still in the loop to override it. However, an agentic system could autonomously change a patient's care plan or adjust a medical device setting based on a faulty or noisy input, potentially leading to immediate and life-threatening consequences. In a similar vein, a hospital logistics agent might also misinterpret an outlier request and reroute critical resources (like ventilators or medications), endangering patient safety in ways a static prediction model never could. Therefore, the system's robustness must be flawless, as a single error could have cascading effects without a human to intervene.

#### D. INTERSECTIONALITY OF TRUSTWORTHINESS DIMENSIONS

Although the expectation is that a TAI system should concurrently satisfy multiple dimensions, these dimensions often interact in complex and sometimes conflicting ways. It is pertinent to establish a balance among the dimensions by understanding their interactions.

##### 1) ACCORDANCE

Certain dimensions of Trustworthy AI (TAI) are mutually reinforcing. Two dimensions are said to be accordant when improvements in one tend to support the other. For instance,

robustness and explainability are synergistic: models trained with techniques like Lipschitz regularization produce clearer saliency maps, whereas Interpretation Regularization (IR) improves both gradient interpretability and adversarial resistance [1]. Similarly, fairness and transparency are tightly coupled, detecting and mitigating bias requires insight into how AI systems generate decisions, supported by explainable outputs and transparent documentation [25], [33], [68].

In LLMs, XAI techniques help uncover privacy and security risks, such as memorized sensitive data or hidden biases within model components. They also play a role in reducing harmful behaviors like toxicity, dishonesty, and hallucination by analyzing the model's internal activations and evaluating the reliability of its outputs [32].

##### 2) CONFLICT

Unlike the accordance, certain dimensions may inherently conflict, making it difficult to satisfy them simultaneously. For high-stake domains like healthcare, it is important to establish trade-offs.

##### a: FAIRNESS VS. PRIVACY

Fairness often requires sensitive group data, raising privacy risks. Privacy-preserving methods like Differentially Private Stochastic Gradient Descent (DP-SGD) may unintentionally harm underrepresented groups through gradient clipping or make them vulnerable to membership inference attacks (MIAs) [30]. For instance, if a hospital trains a model to predict diabetes complications, ensuring fairness requires using sensitive attributes like race and gender, but this raises privacy risks. Meanwhile, applying differential privacy (e.g., DP-SGD) protects patient data but the required noise injection and gradient clipping may disproportionately reduce accuracy for smaller demographic groups, worsening their care. Additionally, adversaries could still exploit model outputs to potentially reveal whether a patient's health record was used in training (MIA).

##### b: FAIRNESS VS. ROBUSTNESS

Reweight to reduce bias can bring samples closer to decision boundaries, increasing vulnerability to adversarial attacks. Conversely, adversarial training may reduce accuracy and exacerbate group disparities. Hybrid frameworks like Fair-Robust Learning attempt joint optimization, albeit with trade-offs [1].

##### c: FAIRNESS VS. EXPLAINABILITY

In fairness, oversampling minority groups or adding fairness constraints during training can shift feature importance, thereby misleading post hoc interpretability tools. Furthermore, explanation fidelity may vary across demographic groups, leading to biased user trust and perception. In some cases, explanations can even obscure underlying biases, giving the illusion of fairness while masking discriminatory behavior [30]. In a cancer risk prediction model for example, oversampling data from minority patients group improves

fairness but shifts feature weights. Post-hoc tools like SHAP can then highlight misleading risk factors, causing doctors to trust explanations differently across groups and potentially masking underlying bias.

#### *d: PRIVACY VS. ROBUSTNESS*

Models trained with DP-SGD weaken robustness due to gradient clipping and added noise, leading to rough decision boundaries [30]. Empirical studies confirm that such models are more easily fooled by adversarial attacks due to the added noise and gradient clipping which creates unstable decision boundaries (e.g., slightly altered X-ray images misclassified as healthy). Conversely, adversarial training used to improve robustness can inadvertently increase susceptibility to privacy attacks like MIAs by enhancing the influence of individual data points.

#### *e: PRIVACY VS. EXPLAINABILITY*

Differential privacy degrades explanation fidelity and stability, whereas local explanation tools like LIME can leak sensitive training information, creating a transparency–privacy paradox [100]. A mental health chatbot trained with differential privacy produces noisy model explanations that reduce clinicians' trust.

#### *f: ROBUSTNESS VS. EXPLAINABILITY*

Adversarial training changes the internal feature representations learned by models. In some domains, like image classification, this leads to more human-interpretable features. However, in other modalities, it may reduce the clarity and fidelity of explanations. Additionally, explanations can be manipulated through adversarial perturbations that leave predictions unchanged but alter the explanation, undermining trust in their reliability.

In a radiology model for example, adversarial training makes image features more robust but shifts internal representations, reducing the clarity of heatmaps used to explain tumor detection. Worse, small adversarial perturbations to image scans can leave predictions unchanged but distort the explanation maps, misleading clinicians and undermining trust.

Addressing these tensions calls for conflict-aware multi-objective optimization and application-specific trade-offs, such as prioritizing robustness in diagnostics and fairness in patient communication.

#### **E. THE UTILITY VS. TAI TRADE-OFF**

Reference [30] examines criticisms of intersectoral TAI, arguing that prioritizing multiple TAI dimensions, such as fairness, robustness, and privacy, may compromise model accuracy and introduce optimization complexity due to conflicting objectives. Critics claim that these trade-offs (e.g. fairness and privacy) can make simultaneous improvement impractical or counterproductive. However, Cresswell contends that utility without deployment is meaningless, but advocates for selective prioritization of TAI goals

through interdisciplinary input and risk-based decision-making, rather than striving for exhaustive optimization.

### **III. METHODS AND TECHNIQUES**

This section outlines the methodological foundations used to evaluate trustworthy AI in healthcare across its core dimensions. We begin by addressing privacy and security, which remain critical in handling sensitive biomedical data and ensuring compliance with strict regulatory frameworks. Next, we examine explainability and interpretability, focusing on techniques that make model reasoning transparent to clinicians and patients. We then turn to robustness and safety, highlighting defenses against adversarial attacks and mechanisms to maintain reliable performance in high-stakes settings. Finally, we consider fairness, emphasizing approaches to identify, measure, and mitigate biases that can propagate inequities across demographic groups. Each subsection details the state-of-the-art methods, their trade-offs, and their implications, providing a structured basis for assessing and advancing trustworthy AI in real-world healthcare applications.

#### **A. PRIVACY AND SECURITY**

Privacy remains a significant challenge in the implementation of deep learning (DL) in biomedical domains, particularly in Mobile Cloud Computing (MCC) and Deep Learning as a Service systems, where data uploads expose sensitive information to risk [118]. Local inference is constrained by resource limitations and proprietary concerns, necessitating Privacy-Preserving Machine Learning (PPML) approaches using secure multi-party computation (SMPC), homomorphic encryption, and garbled circuits. However, these techniques often suffer from latency and accuracy trade-offs.

References [1] and [33], outlined several strategies to improve privacy and security on AI systems. For instance, data anonymization is a fundamental strategy that aims to mask personal identifiers throughout the data collection, publishing, and distribution stages. Methods like k-Anonymity ensure records cannot be distinguished from at least k others, whereas l-Diversity adds intra-group variability, and t-Closeness ensures the distribution of sensitive values remains close to the overall population.

Furthermore, Differential Privacy (DP) is a formal mathematical framework that quantifies privacy loss using parameters  $\epsilon$  (privacy budget) and  $\delta$  (failure probability), offering strong privacy guarantees during data analysis. Reference [30] justified DP as one of the principal frameworks to quantify and limit the privacy leakage of individual data points in machine learning (ML) models. Formally, a probabilistic function  $F$  is said to be  $(\epsilon, \delta)$ -differentially private if, for all neighboring datasets  $D_1$  and  $D_2$  differing in one element, and for all measurable subsets  $S \subseteq \text{Range}(F)$ :

$$\Pr[F(D_1) \in S] \leq e^\epsilon \cdot \Pr[F(D_2) \in S] + \delta$$

The probability of any specific output is nearly the same whether a single data point is included or excluded from

the dataset. This ensures no single data point has a strong influence on the model output, thus bounding the risk of privacy violations. It is important to note that according to [30], Differential privacy in machine learning is commonly achieved via DP-SGD, with the update rule for parameters  $\theta$  at time step  $t$  defined as:

$$\theta_{t+1} = \theta_t - \lambda \left( \frac{1}{|B|} \sum_{i \in B} \text{clip}_C(g_i) + \frac{\sigma C}{|B|} \cdot \xi \right) \quad (2)$$

where:  $\lambda$  is the learning rate,  $B$  is the mini-batch of training examples,  $g_i = \nabla_{\theta} \mathcal{L}(\theta; x_i, y_i)$  is the per-example gradient,  $\text{clip}_C(g_i)$  clips the gradient to norm  $C$ ,  $\sigma$  is the noise multiplier, and  $\xi \sim \mathcal{N}(0, I)$  is standard Gaussian noise.

In addition to empirical metrics such as attack-success rate or F1-score, formal privacy metrics, including  $\epsilon$ -differential privacy budgets,  $\delta$ -failure probability, Rényi-DP [119], and mutual-information leakage, provide mathematically grounded measures for quantifying privacy loss, particularly relevant for federated and cross-institutional healthcare applications [120].

Meanwhile, techniques like machine unlearning have emerged to let models *forget* specific data upon request and without full retraining, aiding regulatory compliance and giving users more control [27], [33], [83]. The key metrics involve measuring how thoroughly the model unlearns data without degrading performance. This is in accordance with the right-to-erasure principle of GDPR. Complementary to this, Secure Computing and Encryption approaches like Secure Multiparty Computation (SMC) and Homomorphic Encryption (HE) enable privacy-preserving computations on encrypted data.

Data Access and Control Mechanisms ensure that sensitive information is only available to authorized users [31], [33]. Evaluations examine the use of access control lists, role-based and attribute-based access control, and how user consent is managed. Systems must also support data deletion requests and restrict model access (e.g., via closed box vs. glass box protocols). Privacy Impact Assessments (PIAs) are structured evaluations to analyze data flows, privacy risks, and compliance with legal standards. These assessments often involve documentation, dataflow diagrams, and checklists, and typically require collaboration between technical teams, legal advisors, and data protection officers. In high-risk applications, such as in healthcare, PIAs may be mandatory under jurisdictional laws [33].

In terms of applications, [121] introduces a framework for cardiovascular disease prediction, integrating data anonymization using Laplace noise on sensitive features, and further achieved patient privacy preservation through a Differential Privacy framework, which yielded promising results comparable to non-privacy models.

Reference [24] show that anonymized medical images can still reveal personal attributes. FL helps mitigate this by enabling collaborative training without data sharing. Reference [118] proposed Privacy-Preserving CNN

(PPCNN), tailored for MCC environments using low-expansion homomorphic encryption (LeHE) and noise masking to secure CNN inference while maintaining performance. The framework operates in three key phases (offline, online convolutional/nonlinear, and fully connected layers) and optimizes linear computations via coefficient-aware packing while reducing communication overhead in nonlinear operations with noise masking. To mitigate FL-related privacy risks, encryption techniques (e.g., DP, SMPC) are used to secure model updates.

Privacy concerns can be addressed through differential privacy to anonymize data by injecting random noise. This can be useful in federated learning and distributed agent settings. DP-SGD and privacy budgets help control information leakage. Also, data minimization and pseudonymization can reduce exposure risks. Secure computing techniques like SMPC, homomorphic encryption, and Trusted Execution Environments (TEE) reduce privacy concerns, particularly in cross-silo applications such as healthcare diagnostics [31].

Reference [37] propose a healthcare AI framework that integrates trust, security, and regulatory compliance. It uses Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) for enhanced diagnostics, employs Advanced Encryption Standard (AES) and multi-factor authentication for data protection, and implements privacy-preserving methods (anonymization/pseudonymization) aligned with GDPR and HIPAA.

Traditional centralized learning faces challenges in data aggregation and privacy, particularly in healthcare [13], [20], [24]. Federated learning mitigates these by enabling decentralized model training across participating clients while sharing only model parameters, thereby reducing data leakage risks. Reference [122] affirm FL's contribution to trustworthy AI, evaluating it in terms of runtime, accuracy across heterogeneous clients, and privacy-preserving measures like Differential Privacy (DP) and adversarial defenses [1], [33]. Although FL significantly improves data security and privacy, vulnerabilities persist, including potential inference attacks from shared updates and threats from malicious clients.

FL is categorized into horizontal (HFL), vertical (VFL), and hybrid types, depending on data partitioning. HFL deals with identical features but different data samples across clients, requiring aggregation methods like FedAVG and FedSGD. Meanwhile, VFL involves shared users but different features, requiring complex aggregation methods like SplitFed. Hybrid FL supports diverse data structures [36]. Data in such distributed systems exist in three states: data-at-rest, data-in-transit, and data-in-use. Despite protections for data-at-rest and in-transit, data-in-use, processed in memory during local training, remains vulnerable to unauthorized access through compromised systems [36]. Local computation on unencrypted data exposes it to leakage risks prior to encryption, both at the client and aggregation server. Consequently, data-in-use is susceptible to memory dumps and other types of unauthorized access or exploitation,

TABLE 6. Privacy and security problems in AI, LLMs, and agentic AI with mitigation techniques.

| Privacy & Security Problems                                                                         | Techniques                                                                                                                                                |
|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sensitive Data Leakage: Models memorizing and revealing PII, medical data, or reconstructing inputs | Federated Learning, Differential Privacy, Homomorphic Encryption, SMPC, anonymization, sanitization, machine unlearning [1], [101]                        |
| Adversarial Attacks: Membership inference, inversion, extraction, poisoning, evasion                | Adversarial training, robust fine-tuning, access controls, model obfuscation, dynamic updates, TEEs, anomaly detection [13], [83], [101]                  |
| Data-in-Use Vulnerabilities: Unencrypted data in memory                                             | Confidential computing, TEEs, encryption during computation [36]                                                                                          |
| Trade-offs: Privacy vs. utility or robustness vs. privacy                                           | SPIN (fairness–privacy balance), tuned DP parameters, balanced robustness–privacy strategies [82], [122]                                                  |
| LLM Risks: Data collection, prompt vulnerabilities, output leakage, fairness–privacy trade-off      | DP-tuned LLMs, SPIN, knowledge unlearning, PII filtering, test-time defenses, prompt preprocessing, NER, Silent Guardian [34], [83]                       |
| LLM Adversarial Exploits: Jailbreaking, prompt injection, model theft, misinformation generation    | Prompt/model-level jailbreak defenses, safety training, RLHF, adversarial fine-tuning, constitutional AI, moderation, watermarking [34], [65], [83]       |
| Agentic AI Privacy: Sensitive data use, persistent memory, inter-agent leakage, real-world capture  | Data minimization, TLS channels, encrypted memory, Privacy-by-Design, consent layers, memory redaction [31], [99]                                         |
| Agentic AI Security: Expanded attack surface, collusion, adversarial interactions, insecure APIs    | LLM-emulated sandboxes (ToolEmu), AgentMonitor, layered defenses, input validation, authentication (PKI, blockchain), goal safety, fail-safes [31], [123] |
| Governance Challenges: Inadequate frameworks, autonomy-control balance                              | AI TRISM, risk management, adaptive control, multi-agent debate, ethical alignment, compliance, traceability [83], [99], [123]                            |

Note: DP: Differential Privacy; NER: Named Entity Recognition; PII: Personally Identifiable Information; PKI: Public Key Infrastructure; SMPC: Secure Multi-Party Computation; SPIN: Suppress Privacy and Fairness Coupled Neurons; TEE: Trusted Execution Environment; TLS: Transport Layer Security; TRISM: Trust, Risk, and Security Management.

posing a significant challenge for secure distributed learning. These vulnerabilities also affect other distributed systems like multi-agent bandits.

Additionally, clients perform local model training using unencrypted data, making them vulnerable during computation. Although model updates are encrypted before transmission, there is a risk of data leakage before encryption occurs on the client side. Similarly, on the aggregation server, data must be decrypted for processing, again rendering it vulnerable in-use. This challenge is not limited to federated learning but extends to other distributed AI paradigms like multi-agent bandits, where reward data-in-use can be compromised even if encrypted at rest or in transit [36].

Reference [36] reviews privacy enhancing technologies (PET) tailored to FL. DP adds noise to gradients, protecting individual data; SMC enables joint computation without revealing private inputs; and HE supports secure computation on encrypted data. These approaches mitigate leakage during training and inference. Evaluations by [122] demonstrate that while PETs may affect training efficiency, they significantly enhance privacy, fairness, and transparency in FL-based AI systems.

Beyond FL, [97] presents an adaptive compression-based framework to enhance privacy preservation in LLMs for sensitive healthcare applications. Recognizing the risk of LLMs leaking private medical information, the framework identifies neurons relevant to privacy and applies adaptive low-rank decomposition (ALRD) to compress these components selectively, reducing the likelihood of data leakage.

To address security and privacy in Agentic AI, a multi-layered defense strategy is required, including encryption, access control, adversarial defenses, and runtime monitoring, all tailored to the unique demands of inter-agent communication. Encryption methods such as SSL/TLS, homomorphic encryption, and secure enclaves protect data exchanged between agents, especially in sensitive domains like healthcare. Access control ensures agents in orchestrated systems follow the principle of least privilege, with Role-Based Access Control (RBAC) and Attribute-Based Access Control (ABAC) policies restricting unauthorized access [31], [33], [97].

Key metrics used to quantify the critical trade-off between privacy guarantees and model utility are detailed in Table 6.

B. EXPLAINABILITY AND TRANSPARENCY

In healthcare and traditional ML and DL, methods such as LIME, SHAP, saliency maps, and counterfactuals help make model decisions transparent. Reference [24] highlights three key metrics for evaluating explanation quality in medical imaging: Insertion (how well important features are identified - higher is better), Deletion (how effectively irrelevant features are removed - lower is better), and Fidelity (how closely explanations match the model’s true decision logic - higher is better). While techniques like GradCAM and LIME are widely used, consistent and rigorous evaluation of their effectiveness is still lacking, making XAI crucial for building clinician trust in privacy-preserving AI systems.

Explanation techniques are often classified as gradient-based and perturbation-based. Gradient-based approaches such as Class Activation Mapping (CAM) and its generalization, Gradient-weighted Class Activation Mapping (Grad-CAM) use backpropagation to highlight important features, particularly in computer vision. Perturbation-based methods like SHAP rely on input modification (e.g., masking or blurring) and game-theoretic principles to assess feature contributions. Counterfactual explanations provide intuitive reasoning by identifying minimal input changes that would alter predictions.

Sometimes, explanation methods can also be categorized by model type, scope, timing, and technique [1], [20]. By model type, explanations are intrinsic (native to transparent models like decision trees and linear regression) or post-hoc (applied to opaque models like deep neural networks). Tools such as LIME and GNNExplainer provide post-hoc interpretability without modifying model internals. By scope, explanations can be local (clarifying individual predictions, e.g., saliency maps for sentiment analysis) or global (describing overall model behavior, e.g., SP-LIME or

class saliency maps) [1], [20]. [20] further distinguishes ex-ante (pre-use) explanations that build initial trust from ex-post (post-decision) explanations that validate decisions.

Reference [20] note that explanation methods span the AI lifecycle: pre-modeling (e.g., data visualization), in-modeling (e.g., using interpretable models), and post-modeling (e.g., proxy models and SHAP, LIME, or rule-based outputs). Meanwhile, concept-based methods like Testing with Concept Activation Vectors (TCAV) [124] and Automated Concept-based Explanations (ACE) [125] explain predictions using human-interpretable concepts instead of raw features, where TCAV measures the influence of predefined concepts (e.g., stripes) and ACE automatically discovers concepts by clustering image patches. They provide semantic, human-friendly explanations, but their reliability depends on the quality of concept definitions or clusters.

Reference [126] argue that traditional XAI methods (e.g., Grad-CAM, SHAP, LIME) are largely post-hoc and do not capture actual reasoning processes. They propose concept-based models that map neural activations to human-understandable concepts and use logic rules to connect them. They proposed Logical Reasoning-based eXplainable Federated Learning (LR-XFL), where local logic rules are aggregated using entropy-based models and beam search to enhance interpretability without accessing raw data. The method dynamically filters low-accuracy or biased contributions, improving transparency. However, LR-XFL faces limitations like unclear bias handling and suboptimal rule selection due to greedy search strategies. Therefore, integrating FL and XAI provides a balanced framework where FL preserves data privacy and XAI ensures transparency. Evaluating the effectiveness of XAI not only reveals hidden biases but also enhances stakeholder trust and improves the reliability of AI-assisted.

Reference [91] developed an explainable electrocardiogram (ECG) classification framework using Federated Transfer Learning (FTL). Grad-CAM highlights signal regions (e.g., P-wave, QRS complex), supporting clinical interpretability while maintaining privacy in a federated environment. This work exemplifies the synergy between privacy-preserving and explainable AI in real-world healthcare applications.

In [127], showing AI confidence scores and local explanations had differing effects on human trust calibration and decision accuracy in high-stakes AI-assisted tasks, emphasizing the importance of users forming accurate mental models of AI error boundaries. The study finds that confidence scores significantly support trust calibration by helping users decide when to rely on AI predictions, even without seeing the predictions themselves. This coincides with the findings of [89]. In contrast, local explanations (e.g., SHAP-based) were ineffective in influencing trust, possibly due to their ambiguity or cognitive load. The work focuses on scenarios where human and AI performance are comparable.

XLR-KGDD [128], is a framework for explainable multimodal disease diagnosis that integrates a patient

knowledge graph, LLM, and RAG. Unlike traditional XAI methods, it produces evidence-based explanations grounded in physician-endorsed medical guidelines, ensuring generalizability across datasets. The system combines multimodal data, including symptoms, radiology images, and demographics, into a patient knowledge graph to provide the LLM with a rich, integrated context for diagnosis. It also employs parameter-efficient fine-tuning (e.g., QLoRA) to enable local, resource-efficient deployment in clinical settings.

Reference [33] outline five strategies for evaluating transparency: computational explanation, human-based explanation, openness, documentation review, and input/output transparency. Computational metrics assess faithfulness (alignment with actual importance), robustness (stability under perturbation), and complexity. Counterfactuals are assessed via proximity, sparsity, plausibility, and diversity. Human-based evaluation measures comprehensibility, confidence, satisfaction, and task performance through user studies. Openness refers to access to source code, training data, and design documentation. Documentation reviews use tools like model cards and datasheets to explain datasets, algorithms, and ethical considerations. Input/output transparency assesses how well users understand system assumptions, expected outputs, and potential limitations, typically evaluated via interviews or usability testing.

In LLMs, latent representations contain rich, though polysemantic information, and hence representations are not inherently interpretable. Non-training-based methods like Logit Lens interpret hidden states by projecting them onto the output embedding space to reveal likely word activations, although distribution shifts between layers pose challenges. Meanwhile, sparse Autoencoders (SAEs) offer an unsupervised approach to extract monosemantic concept vectors by enforcing sparsity in activations, and only the top-K components are active. However, recent studies show that input-based interpretations may not reflect how these vectors influence output behavior, prompting a shift toward output-based interpretations to better connect learned concepts with model responses [32].

Reference [129] proposes a framework for explainable zero-shot medical image classification by integrating a vision language Model (CLIP) with ChatGPT. To address the scarcity of labeled medical data and enhance interpretability, the framework uses ChatGPT to generate symptom-based textual descriptions that guide CLIP's visual attention, mimicking expert diagnostic reasoning. The method improves explainability through attention maps and textual cues while mitigating LLM hallucinations via carefully designed prompts grounded in medical literature. Statistical analysis confirms that the diagnostic accuracy aligns with high textual-image similarity, and the study demonstrates the promising role of LLMs in medical AI.

Explainability and transparency are crucial for building user trust in multi-agent AI systems, as outcomes result from complex interactions rather than a single decision. The challenge lies in interpreting not just individual agent choices but

TABLE 7. Explanation quality metrics.

| Metric                | Definition/Interpretation                                                                               |
|-----------------------|---------------------------------------------------------------------------------------------------------|
| Fidelity              | Agreement between explanation and model behavior; Compare predictions with/without highlighted features |
| Consistency           | Stability of importance scores across similar inputs; Measure variance of attributions                  |
| Continuity            | Similar inputs yield similar explanations; $\ \text{Exp}(x) - \text{Exp}(x')\ $ for close $x, x'$       |
| Selectivity           | Focus on most influential features only; Sparsity of non-zero attributions                              |
| Robustness            | Explanation stability under perturbation; Add noise to input; measure explanation change                |
| Completeness          | Full coverage of decision factors; Verify all relevant features included                                |
| Insertion Score       | Performance when adding features by importance; Higher is better; tests positive impact                 |
| Deletion Score        | Performance when removing features by importance; Lower is better; confirms feature necessity           |
| Localization Accuracy | IoU between explanation heatmap and ground truth; Measures spatial alignment in imaging                 |
| Pointing Game         | Hits when max attention falls in object region; Binary localization accuracy                            |
| Attribution AUC       | Area under precision-recall for pixel attributions; Overall explanation quality                         |

Note: AUC: Area Under the Curve; IoU: Intersection over Union.

also the dynamics of their collaboration. To achieve this, [31] explored several methods. For instance, feature attribution techniques like LIME and SHAP can analyze which inputs most influence an agent’s output. Counterfactual analysis determines how a system’s behavior would change if an agent’s actions were altered, helping to isolate each agent’s role.

Some key techniques for making an agent’s internal decision-making process understandable to humans include reasoning traceability, where methods like CoT logging and decision provenance graphs create a clear audit trail of an agent’s thought process and collaborations. Comprehensive logging ensures that all actions and inter-agent messages are captured, with immutable logs providing tamper-proof records. The generative power of Large Language Models can be used to create human-friendly explanations, where a dedicated explainer agent translates complex logs into simple narratives [31], [99]. However, [117] demonstrated that CoT reasoning in agentic AI was weakly correlated with answer quality, often producing verbose but misleading explanations.

Finally, a trade-off exists between transparency and efficiency. While interpretable models (e.g. decision trees) are transparent but often less accurate, deep learning

models offer high performance with limited interpretability, a dilemma known as the accuracy-interpretability trade-off [100]. In healthcare, transparency often demands detailed documentation of data, training, and decision logic, which can increase computational burden.

A comprehensive overview of objective metrics used to evaluate different dimensions of explanation quality, such as fidelity and consistency, is compiled in Table 7.

C. ROBUSTNESS AND SAFETY

To advance adversarial robustness, [29] highlight several tools and frameworks designed for evaluating and enhancing AI model resilience. CleverHans (Google) and Advtorch (PyTorch) enable benchmarking against various attack methods, whereas ART (Adversarial Robustness Toolbox) supports multi-modal robustness testing across text, image, audio, and video. Foolbox (Bethge Lab) focuses on generating adversarial examples, and DeepCert provides formal robustness guarantees using verifiable neural network properties. Reluplex applies SMT solvers to validate ReLU-based models. Robust VAE techniques strengthen generative models using adversarial training and regularization. Additional libraries such as DeepRobust and RobustBench support image/graph robustness analysis and standardized evaluations with pre-trained robust models [1], [29].

Robustness evaluation typically involves attack algorithms like Fast Gradient Sign Method (FGSM), Projected Gradient Descent (PGD), and DeepFool, using metrics such as attack success rate, fooling rate, empirical robustness, and the CLEVER score. Certified robustness methods further offer mathematical assurances of stability within perturbation bounds [29], [33]. Additionally, [89] evaluated Mahalanobis, Gram matrices, and energy-based scoring across diverse health modalities and showed that standard models often fail under OOD inputs. The study focused on image (skin cancer), audio (lung sounds), and motion (Parkinson’s disease) data. Their proposed confidence score improved user trust and understanding. Common metrics employed to quantify a model’s performance under distribution shift and for OOD detection are presented in Table 9.

In GPT models, instruction tuning and RLHF reduce toxicity and improve resilience to benign prompts, contributing to general robustness. However, the susceptibility of the model to misleading prompts suggests limitations in robustness against adversarial misuse [104]. Multiple studies, including [130], have explored the use of knowledge editing to reduce toxicity in LLMs.

To protect against threats like prompt injection, spoofing, and impersonation in multi-agentic AI systems, [31] advocates a deep defense approach like prompt hygiene, where agents defensively sanitize and filter input, and apply guardrails to refuse suspicious instructions. Trust scores or reputation systems where agents verify each other’s output, and training agents with adversarial robustness can improve their resistance to malicious inputs.

Reference [131] proposes a Human-AI Collaboration (HAIC) guided deferral system to make LLMs safer and more practical for healthcare. The system identifies uncertain predictions and defer them to a human expert, providing intelligent guidance to improve the expert's decision-making. This approach uses open-source LLMs that are efficient enough to run on standard hardware, addressing concerns about privacy and accessibility while remaining competitive.

Reference [132] introduced SALAD-Bench, a pioneering safety benchmark that evaluates LLMs across multiple types of task and jailbreak attack scenarios. While comprehensive, it does not examine how different types of tests relate to each other or how prompt engineering strategies influence safety outcomes. Building on these gaps, SG-Bench [133] is designed to systematically assess the generalization of LLM safety by spanning various evaluation formats, including open-ended generation, multiple choice tasks, and safety judgments, while incorporating a variety of jailbreak attacks and prompt engineering techniques into its testing framework.

Detecting hallucinations typically requires ground truth references or labeled datasets, which are expensive and domain-specific. As a result, [134] introduces HaloScope, an unsupervised framework for hallucination detection in LLMs that leverages unlabeled model-generated text, removing the need for human annotations or external references. The framework identifies a latent subspace within LLM embeddings to estimate the likelihood of hallucination and trains a binary truthfulness classifier based on these estimates. It also evaluates internal consistency through self-verification and follow-up probing, enhancing detection accuracy.

Mitigation pipelines increasingly incorporate retrieval-augmented grounding, self-consistency sampling, answer-cross-checking, constrained decoding, and rule-based verification layers [135], [136], [137], [138], [139], [140]. These mechanisms reduce unsupported claims and improve factuality in clinical content generation.

Reference [1] outlined several defense mechanisms such as adversarial training, which incorporates adversarial examples into the training set. Although it improves robustness, it comes with significant trade-offs like slower training and reduced accuracy on clean data. Methods like gradient approximation aim to alleviate these issues. Certified defenses take a more formal approach by providing provable guarantees against norm-bounded perturbations, although at the cost of increased computation [1], [26], [36]. Unlike empirical methods, these aim to mathematically guarantee that no adversarial input exists within a certain range. Techniques like randomized smoothing aggregate predictions from perturbed inputs to produce a more stable outcome. Detection-based defenses, which aim to identify adversarial examples during inference, have had mixed results. While some rely on statistical tests, others monitor user query

**TABLE 8. Robustness and safety problems with mitigation techniques AI.**

| Problems                                                                                                                                                  | Techniques / Solutions                                                                                                                                                                                                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Vulnerability to Imperceptible Perturbations</b> – ML models like CNNs, GNNs, RNNs and LLMs can be fooled by tiny adversarial changes [1], [27], [101] | Adversarial training, certified defense (Randomized Smoothing), adversarial example detection, robust feature engineering [26], [33], [36], [83], [134]                                                               |
| <b>Conflicting Requirements / Trade-offs</b> – Robustness vs fairness/accuracy/privacy [64], [82]                                                         | Joint optimization, responsibility–utility co-design, multidisciplinary approach, neuro-symbolic methods [26], [64], [82], [84]                                                                                       |
| <b>Immature / Fragile Approaches</b> – Many robustness/explainability methods fragile to perturbations or hard to verify [1], [26], [30], [83]            | Hybrid model design, iterative robustness evaluation, faithfulness testing, controlled perturbation testing [31], [33], [100]                                                                                         |
| <b>Quantification Challenges</b> – Hard to measure trustworthiness metrics [26], [32], [33]                                                               | Comprehensive evaluation frameworks, automated tools (Cleverhans [142], AdvTorch [143], DeepRobust [144], RobustBench [145]), SALAD-Bench [132]                                                                       |
| <b>Limited Generalizability</b> – Poor adaptation to unseen domains or shifting data [26], [27], [64], [97]                                               | Domain adaptation, continual learning, synthetic data generation [26], [32], [65], [95]                                                                                                                               |
| <b>Data Risks</b> – Poisoned/skewed data, privacy threats [1], [36]                                                                                       | Data validation/cleansing, anonymization, encryption, differential privacy, federated learning, synthetic data [20], [26], [31], [33], [36], [83]                                                                     |
| <b>LLM Hallucinations / Unsafe Content</b> – Non-factual, toxic, biased, or illegal outputs [23], [32], [65]                                              | Guardrails (input/output filters, neural classifiers), RLHF, instruction tuning, RAG with external KBs, safety decoding, self-reminder prompts, goal prioritization, multilingual safety [23], [27], [32], [64], [65] |
| <b>LLM Prompt Attacks</b> – Jailbreaking, injection, cross-lingual exploits [64], [65], [83]                                                              | Robust prompt optimization (RPO), perplexity-based filtering, prompt hygiene, classifier-based detection, sandbox testing [27], [64], [83]                                                                            |
| <b>LLM Fine-tuning Vulnerabilities</b> – Small-data tuning can harm safeguards, leak PII [64], [82]                                                       | Safety-aware fine-tuning, DP-tuned models, fine-tuning audits, machine unlearning [32], [64], [83], [97]                                                                                                              |
| <b>Agentic AI Coordination / Error Propagation</b> – Multi-agent orchestration failures, emergent misbehavior [31], [99], [123]                           | Goal safety protocols, fail-safes, bounded autonomy, TRISM framework, Component Synergy Score (CSS), Tool Utilization Efficacy (TUE) [26], [31], [99]                                                                 |
| <b>Agentic AI Security / Malicious Use</b> – Collusion, prompt injection, dual-use misuse [31], [83], [99], [123]                                         | Prompt hygiene, sandboxing, proactive risk assessment, blockchain/PKI-based identity, secure communications channels [31], [99], [123], [146]                                                                         |
| <b>Agent Motivation Drift / Physical Risks</b> – Goal divergence, embodied AI hazards [27], [65], [123]                                                   | Bounded autonomy, environmental adaptation, regulatory compliance, physical safety standards, HITL oversight [31], [99], [123]                                                                                        |

Note: CNN: Convolutional Neural Network; CSS: Component Synergy Score; DP: Differential Privacy; GNN: Graph Neural Network; HITL: Human-in-the-Loop; KB: Knowledge Base; LLM: Large Language Model; RAG: Retrieval-Augmented Generation; RLHF: Reinforcement Learning from Human Feedback; RNN: Recurrent Neural Network; RPO: Robust Prompt Optimization; TRISM: Trust, Risk, and Security Management; TUE: Tool Utilization Efficacy.

history for suspicious patterns. However, distinguishing adversarial from natural samples remains a significant challenge [1].

Equation 3 outlines a critical update rule for model parameters ( $\theta$ ) within an adversarial training framework:

$$\theta_{t+1} = \theta_t - \lambda \frac{1}{|B|} \sum_{x_i \in B} \nabla L(\theta_t; x_i^\dagger(\theta_t, x_i, y_i), y_i) \quad (3)$$

This formulation adapts conventional mini-batch gradient descent by integrating the concept of adversarial examples directly into the optimization process. Here,  $\theta_t$  denotes the model parameters at iteration  $t$ ,  $\lambda$  is the learning rate, and  $|B|$  represents the size of the mini-batch. Crucially, the gradient

**TABLE 9. Distribution shift and OOD detection metrics.**

| Metric/Method                | Description, Interpretation                                                      |
|------------------------------|----------------------------------------------------------------------------------|
| <i>Distribution Shift</i>    |                                                                                  |
| Accuracy Drop                | $acc_{train\_dist} - acc_{test\_dist}$ , Performance degradation measure         |
| Regret                       | $Oracle_{acc} - Model_{acc}$ , Loss vs. optimal on shifted data                  |
| Domain Divergence            | KL-divergence, Distance between distributions                                    |
| <i>OOD Detection Methods</i> |                                                                                  |
| Max Softmax Prob.            | $\max_y P(y x)$ , Baseline; threshold at 95th percentile                         |
| ODIN Score                   | Temperature-scaled softmax with perturbation, Improved separation of in/out-dist |
| Mahalanobis Distance         | Distance from feature space center, Detect distributional anomalies              |
| Energy Score                 | $-T \cdot \log \sum_y e^{f_y(x)/T}$ , Lower energy = more likely OOD             |
| AUROC                        | Area under ROC for OOD detection, $> 0.9$ desirable for safety-critical apps     |
| FPR@95TPR                    | False positive rate at 95% true positive, Standard OOD benchmark metric          |

Note: AUROC: Area Under the Receiver Operating Characteristic; FPR: False Positive Rate; KL: Kullback-Leibler; OOD: Out-of-Distribution; TPR: True Positive Rate.

**TABLE 10. Uncertainty quantification metrics.**

| Metric                  | Usage                                          |
|-------------------------|------------------------------------------------|
| Negative Log-Likelihood | Probabilistic accuracy; lower is better        |
| Brier Score             | Mean squared error of probabilities            |
| ECE                     | Avg. gap between confidence & accuracy         |
| MCE                     | Worst-case calibration gap                     |
| AUSE                    | Performance with uncertain predictions removed |
| MI                      | Epistemic uncertainty measure                  |

Note: AUSE: Area Under the Sparsification Error; ECE: Expected Calibration Error; MCE: Maximum Calibration Error; MI: Mutual Information.

of the loss function,  $L$ , is computed not with respect to the original input  $x_i$ , but with respect to an adversarially perturbed input,  $x^\dagger(\theta_t, x_i, y_i)$ . This perturbation, generated to maximally increase the loss for the current model parameters  $\theta_t$  while remaining perceptually close to  $x_i$ , forces the model to learn from its vulnerabilities.

Reference [36] focus on the robustness of distributed AI - a class of distributed learning algorithms that enables

multiple clients to jointly train a global machine learning model while keeping their data private and local. Robustness in this case would imply that these systems maintain consistent and accurate performance despite being exposed to various challenges, such as irregular or non-IID data, security vulnerabilities, privacy violations, and biases. They categorize responses as (1) auto-repair without detection, (2) detection without repair, and (3) combined detection-repair systems. Just-in-time detection and human-in-the-loop mechanisms are emphasized, although challenges remain in repairing detected anomalies or ensuring detection precision. Despite the countermeasures summarized in Table 8, a fair and comprehensive safety evaluation can still be marred by major gaps like the shortage of non-English datasets, outdated subset of datasets, and the absence of naturalistic safety evaluations [141].

#### D. FAIRNESS

For fairness, methods include modifying word embeddings, counterfactual data augmentation, resampling, reweighting, adversarial debiasing, and specific post-processing algorithms like Population Sensitivity-Guided Threshold Adjustment (PSTA) and Reject Option Classification (ROC).

Reference [66] evaluates post-processing debiasing methods for mitigating bias in clinical machine learning classifiers. These approaches are advantageous as they preserve privacy and can be applied to pre-trained models without retraining. The study examines the balance between fairness and accuracy under fairness definitions such as Equal Opportunity (EOp) and Equalized Odds (EOd). Seven state-of-the-art methods are tested on eight datasets (four medical, four non-medical) and synthetic data: Calibrated Equalized Odds Post-Processing (CPP), Equalized Odds Post-Processing (EPP), Reject Option Classification (ROC), Population Sensitivity-Guided Threshold Adjustment (PSTA), Fairness-Confusion Tensor (FACT), Group-Specific Threshold Adaptation for Fair Classification (GSTAR), and Multiaccuracy Boost (MAB). Effectiveness is measured using Accuracy, Balanced Accuracy, F1-score, and fairness metrics (EOp, EOd). The Fairea framework is employed to visualize fairness-performance trade-offs, where a debiasing method is considered effective only if it lies above the baseline in the win-win or good trade-off regions, with the added requirement of maintaining at least 90% of performance in high-stakes medical applications. [12] defines the core group fairness metrics essential for assessing equitable treatment in healthcare AI systems.

Several open-source toolkits and repositories developed to support fairness in AI are highlighted in [1] and [20]. For instance, Responsibly offers datasets and metrics for evaluating bias in classification and NLP tasks. FairTest introduces the Unwarranted Associations framework to detect unfair user treatment. AIF360 provides widely used datasets and debiasing algorithms for binary classification. Aequitas serves as an audit toolkit to assess

**TABLE 11. Bias and fairness problems in AI healthcare with mitigation techniques.**

| Problems                                                                                                                                                                                   | Techniques / Solutions                                                                                                                                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Data Bias:</b> Representation gaps, societal bias, annotation bias, mismatched demographics, confounding variables, privacy–fairness conflicts in ML/LLM training data [66], [67], [95] | Diverse/representative datasets, counterfactual data augmentation (CDA), data cleaning, reweighting/resampling, blinding methods, ethical data principles [15], [64], [67], [83], [95]                                                                     |
| <b>Algorithmic / Model Bias:</b> Objective functions, biased estimators, design choices, FL+DP aggravating disparities [20], [36]                                                          | Fairness-aware algorithms, constraint-based optimization, regularization penalties, adversarial learning for fairness, debiasing embeddings/attention heads, SPIN neuron suppression, instruction tuning with CoT rationales [15], [64], [65], [82], [100] |
| <b>Evaluation Bias:</b> Inappropriate metrics cause skewed performance (e.g., facial recognition bias) [20], [65]                                                                          | Use fairness metrics (demographic parity, equal opportunity, equalized odds), auditing tools (AlF360, Aequitas, Fairlearn, DiscrimEval), benchmarks (SALAD-Bench) [33], [65], [66], [82], [95]                                                             |
| <b>Bias in LLMs:</b> Propagation of stereotypes, cultural/gender/racial bias, fairness–privacy trade-off under SFT [34], [65], [67]                                                        | SPIN for fairness–privacy co-optimization, debiasing fine-tuning, fairness-aware prompt design, controlled data curation [64], [82], [101]                                                                                                                 |
| <b>Health Disparities:</b> Unequal diagnosis/treatment across subgroups [15], [66], [68]                                                                                                   | Representative datasets, targeted augmentation, group-specific threshold adjustment, HITL oversight [15], [23], [66], [68]                                                                                                                                 |
| <b>Resource Misallocation / Stifled Innovation:</b> Skewed prioritization towards well-represented groups [15]                                                                             | Multidisciplinary fairness governance, equitable policy design, continual retraining with diverse data [15], [31], [64], [65]                                                                                                                              |
| <b>Fairness–Privacy/Robustness Trade-offs:</b> Optimizing one may harm another [30], [82]                                                                                                  | Joint optimization frameworks, Pareto-based post-processing (PSTA, GSTAR), SPIN balancing fairness–privacy, robustness-aware fairness training [30], [66], [82]                                                                                            |
| <b>Post-processing Needs:</b> Bias mitigation without retraining [15], [66]                                                                                                                | Threshold adjustment, ROC, PSTA, FACT, GSTAR, MAB, EPP, CPP [1], [15], [20]                                                                                                                                                                                |
| <b>Monitoring and Evaluation Gaps:</b> Lack of continuous fairness checks [32], [132]                                                                                                      | Statistical monitoring, user feedback loops, continuous refinement, clinical validation, Fairea framework for fairness–performance trade-offs [15], [64], [95], [147]                                                                                      |
| <b>Explainability Gaps:</b> Hidden bias mechanisms in closed box models [1], [32]                                                                                                          | Model-agnostic XAI (LIME, SHAP), saliency maps, counterfactuals, transparency tools to reveal bias causes [15], [20], [30], [128]                                                                                                                          |

*Note:* CDA: Counterfactual Data Augmentation; CoT: Chain-of-Thought; CPP: Calibrated Equalized Odds Post-Processing; DP: Differential Privacy; EPP: Equalized Odds Post-Processing; FACT: Fairness-Confusion Tensor; FL: Federated Learning; GSTAR: Group-Specific Threshold Adaptation for Fair Classification; HITL: Human-in-the-Loop; LIME: Local Interpretable Model-agnostic Explanations; MAB: Multiaccuracy Boost; PSTA: Population Sensitivity-Guided Threshold Adjustment; ROC: Reject Option Classification; SHAP: SHapley Additive exPlanations; SPIN: Suppress Privacy and Fairness Coupled Neurons; XAI: Explainable AI.

model fairness across demographic groups, and Fairness Measurements includes datasets and code for quantitatively evaluating discrimination in classification and ranking systems.

In-context learning can mitigate bias in LLMs by employing demographically balanced few-shot examples, which prompts models to reject biased statements under neutral conditions. However, misleading system prompts remain a challenge, as they can still induce agreement with biased content, highlighting the need for careful prompt design alongside data balancing techniques [104].

**TABLE 12. Core fairness metrics for healthcare AI.**

| Metric                  | Interpretation                                                    |
|-------------------------|-------------------------------------------------------------------|
| Demographic Parity      | Equal positive classification rates across groups $a$ and $b$     |
| Equal Opportunity (EOp) | Equal True Positive Rates (sensitivity) across groups             |
| Equalized Odds (EOd)    | Equal TPR and FPR ensures equitable correct/incorrect predictions |
| Accuracy Disparity      | Maximum accuracy difference between demographic groups            |
| Disparate Impact        | Ratio of positive outcomes; $\geq 0.8$ often required             |
| Predictive Parity       | Equal precision across groups                                     |
| Counterfactual Fairness | Predictions unchanged if sensitive attribute altered              |

*Note:* FPR: False Positive Rate; TPR: True Positive Rate.

### E. PRIVACY-FAIRNESS TRADEOFF

Reference [82] identifies a trade-off in LLMs where improving privacy awareness through Supervised Fine-Tuning reduces fairness awareness, due to shared (coupled) neurons influencing both. To address this, the authors propose Suppress the Privacy and fairness coupled Neurons (SPIN), a training-free method that suppresses these coupled neurons, effectively decoupling fairness and privacy representations. SPIN improves both fairness and privacy simultaneously across multiple models without harming general capabilities and remains effective even with limited or malicious data.

### F. KNOWLEDGE EDITING IN LLMs

Reference [130] argues for knowledge editing, a post-hoc modification of internal representations of LLM to adjust particular facts or behaviors without retraining them from scratch. This is necessary because LLMs store vast factual, commonsense, and domain-specific knowledge, which can contain errors, produce harmful outputs or over time become outdated. Since traditional retraining is costly and inefficient, knowledge editing offers a lightweight approach to insert, modify, or erase specific knowledge while preserving the general performance of the model. They propose KnowEdit, a benchmark to test insertion, modification, and erasure across factual, sentiment, and privacy-related tasks, evaluated using edit success, portability, locality, and fluency.

Knowledge editing can be useful in ensuring trustworthiness of LLMs. For instance, [148] proposes REMEDI, which uses knowledge editing to reduce gender bias in LLMs, [149] uses knowledge unlearning to reduce privacy risks for LMs post hoc by performing gradient ascent on target token sequences without affecting the performance of the model. Knowledge sanitization can also be used to mitigate extraction attacks and minimize hallucinations, thereby preventing the leakage of sensitive information [150].

### G. STAKEHOLDERS' PERSPECTIVE

Reference [27] identifies six primary stakeholder groups in AI governance: governments, regulators, AI system developers, AI service providers, academics, and the public. Governments play a central role by enacting legislation, setting standards, and coordinating across stakeholders. Regulators, under government oversight, develop policies, audit AI systems, and ensure compliance with legal and ethical norms. Developers build AI systems in coordination with service providers and researchers, whereas providers operationalize these systems and integrate user feedback. Academics contribute foundational knowledge and risk assessments that guide policy and technical design. Consumers provide critical feedback that informs regulatory updates and system improvement.

Reference [28] explores stakeholder perspectives in the context of AI deployment for intrapartum care, revealing that trust in AI often reflects trust in the institutions behind it. Participants viewed academic researchers as more aligned with public interest than private companies, raising ethical concerns over profit-driven development. Ensuring trustworthy collaborations between public institutions and private actors requires transparency, data safeguards, and ethical governance frameworks. The stakeholders also emphasized the importance of human mediation in AI-informed healthcare decisions. Concerns were raised about fairness and equity, particularly when AI systems perform unequally across demographic groups. This led to ethical questions about whether AI tools should be deployed only where performance is adequate, and how their use might unintentionally exacerbate existing health disparities.

### H. EVALUATION

Reference [25] developed a patient-centric evaluation framework - Trustworthy AI Scoring System (TAI-SS) to assess the ethical, privacy, and reliability dimensions of AI systems in healthcare. Built through a rigorous combination of theoretical research and expert consultation, TAI-SS draws on existing standards like the EU Ethics Guidelines, HIPAA, and GDPR. It evaluates AI across ten key domains, including ethical adherence, data privacy, reliability, transparency, security, fairness, and continual monitoring, and then assigns systems a trustworthiness classification from Class I (very high) to Class IV (moderate). The scoring is dynamic and adaptable, incorporating real-world performance, user feedback, and evolving risk profiles.

### I. OPERATIONALIZING TAI: VALIDATION PRACTICES AND BENCHMARKS

To implement the high-level principles of TAI, concrete validation practices and benchmarks are required to move from abstract goals to deployable, verifiable systems. This requires continuous post-deployment monitoring, health outcome tracking, and continuous auditing [25], [95]. These processes are actualized through robust

human-in-the-loop mechanisms, ensuring a clinician-in-loop is maintained to validate or override AI-generated outputs. Furthermore, a trustworthy model must know its own limits, which is implemented through uncertainty evaluation and uncertainty quantification [30], [33], [64]. The goal is to avoid miscalibration, where a model expresses unwarranted confidence by using metrics like ECE, thereby addressing the critical lack of uncertainty expression. Technically, this is enforced using deployment guardrails, such as modular input and output filters that intercept suspicious instructions or harmful content [27], [65]. Finally, to ensure models are safe, fair, and robust, evaluation relies on standardized benchmarks. For robustness, this includes toolkits like CleverHans [142], DeepRobust [144], RobustBench [145], and safety-specific tests like SALAD-Bench [132], while fairness is audited using tools like AIF360 [151] and Aequitas [152].

## IV. RESEARCH GAPS AND FUTURE DIRECTIONS

Following a comprehensive review of the current body of literature, the central aim of this section is to critically examine and articulate key research gaps that hinder the advancement of Trustworthy AI in healthcare and other domains where its application presents heightened ethical and societal challenges. The identification of these gaps serves not only to highlight limitations in the existing scholarship, but also to direct future research toward areas requiring deeper exploration and innovation.

### A. SOCIO-TECHNICAL AND ETHICAL CHALLENGES

#### 1) DIVERSITY AND REPRESENTATIVENESS IN DATASETS

Insufficient diversity in training datasets remains a fundamental barrier to trustworthy AI in healthcare. When datasets disproportionately represent certain populations, such as patients from specific age ranges, genders, ethnicities, or socioeconomic groups, AI models tend to overfit to majority patterns while underperforming for minority subgroups, resulting in misdiagnosis, delayed treatment, or inappropriate recommendations for underrepresented patients [20], [89].

To address these gaps, future research should prioritize methods that augment minority data classes through techniques such as counterfactual data augmentation, synthetic data generation (e.g., GANs, diffusion models), and adaptive resampling. Beyond technical fixes, healthcare institutions must be incentivized to collect inclusive and representative datasets, ensuring that diverse demographics, rare conditions, and intersectional attributes (e.g., age  $\times$  gender  $\times$  ethnicity) are adequately captured.

Equally important is post-deployment monitoring, since even well-designed models can drift over time as population distributions shift or new biases emerge. Continuous auditing, fairness-aware performance dashboards, and clinician feedback loops are critical to detect and correct inequities in real-world use. Such monitoring should not only evaluate

accuracy but also disaggregated fairness metrics across subgroups to ensure equitable outcomes. Ultimately, achieving diversity and representativeness is not a one-time technical intervention but an ongoing socio-technical commitment involving dataset governance, stakeholder oversight, and adaptive model refinement.

## 2) USER TRUST VERSUS AI TRUSTWORTHINESS

The interplay between user trust and the actual trustworthiness of AI systems is critical for successful adoption but also poses societal and economic risks when misaligned [34]. Users often rely on available information, which can skew perceptions, leading to over-trust or under-trust. For instance, positive reports can improve acceptance. Over-trust arises when users overestimate AI capabilities or its trustworthiness, leading to misuse (e.g., delegating critical decisions without oversight) or abuse (e.g., unethical data exploitation). Conversely, under-trust results in disuse, even when AI systems are accurate and reliable. This may stem from transparency gaps or unfamiliarity with AI technologies, limiting innovation and efficiency. For instance, patients may reject AI-based diagnostics, and older adults may avoid AI-powered fraud detection in banking despite its proven effectiveness [34]. To address this kind of issues in healthcare, explainability, transparency, and ethical governance are vital to bridging the gap between perceived and actual system reliability. Ensuring alignment and clear safeguards against errors or biases foster responsible adoption and reinforces trust grounded in human values and societal expectations. Hence, more research is needed to develop human-centric explainability approaches tailored to diverse user profiles and needs [1], [26], [28].

Reference [21] views TAI in healthcare as a socio-technical challenge, emphasizing that organizational accountability shapes trust more than individual human-AI interactions. Their framework advocates for AI systems to provide recipient-designed accounts aligning with human responsibility practices, supporting governance that assigns human accountability and promotes regulatory oversight [20].

## 3) ORGANIZATIONAL ACCOUNTABILITY

Reference [21] examines the practical challenges of TAI in healthcare applications and argue that achieving TAI in healthcare requires more than technical explainability; it must include organizational accountability. Based on ethnographic studies of breast cancer screening and treatment planning, they emphasize the importance of AI systems providing explanations aligned with institutional workflows and user needs, thereby enabling system-level auditing and monitoring.

## 4) EXTENDED INFORMED CONSENT

One key challenge for trustworthy AI in healthcare is establishing a system of extended informed consent. Unlike traditional consent, which focuses mainly on medical procedures and data usage, patients must now also understand

the role of AI in diagnosis, treatment, and monitoring [15], [25]. This requires providing clear and accessible information about how sensitive data is collected, processed, and shared, the degree to which AI influences clinical decisions, and the risks, biases, and limitations of the technology. Without this level of transparency, patients cannot make truly informed choices about their care.

## B. OPEN PROBLEMS ACROSS KEY DIMENSIONS OF TAI

### 1) BIAS AND FAIRNESS IN SPECIFIC APPLICATION CONTEXT

Reference [20] emphasizes the need for fairness frameworks tailored to specific application contexts, improved testing strategies for probabilistic and non-reproducible systems, and policies that address accountability and privacy in a context-sensitive manner. They also advocate for improved methods to evaluate user comprehension of explanations, as well as mechanisms to manage expectations and unravel bias from the design phase.

Meanwhile, [66] proposes adaptive debiasing frameworks that adjust dynamically to dataset properties to maintain fairness. They recommend hybrid strategies to mitigate fairness-performance trade-offs and highlight the need for multi-attribute fairness optimization to address intersectional biases, particularly in clinical datasets. While their work focuses on structured tabular data, future research should expand to include modalities such as medical imaging, audio, EHRs, and genomics.

The long-term impact of AI systems on healthcare fairness remains insufficiently studied, as most current research concentrates on immediate algorithmic biases and short-term performance outcomes. However, it is essential to recognize that initial biases, if left unchecked, can compound over time, potentially worsening existing health disparities. To address this, future research must prioritize longitudinal evaluation by continuously monitoring AI systems after deployment [15], [67]. Such monitoring should assess the differential impacts on various demographic groups at regular intervals, allowing for the detection of emerging inequities. Reference [15] advocate for adaptive feedback loops, enabling clinicians to report disparities and iteratively refine AI models.

### 2) THEORIZATION IN EXPLAINABILITY AND INTERPRETABILITY (XAI)

The closed box nature of AI models hampers trust, especially in critical domains like healthcare. There is a need for stakeholder-specific, human-understandable explanations [1], [90], [101]. Key challenges include evaluating explanation accuracy, ensuring robustness, and supporting non-technical users. Mechanistic interpretability methods often lack generalizability and require formal, empirical validation. Future research should develop general theories to explain the underlying mechanisms of AI models, analyze a broader range of model parameters, uncover reasoning capabilities, and examine how factual knowledge is represented within MLP layers [27].

Future work should develop foundational theories to explain model behavior, analyze model parameters, and build explainable federated learning with logical reasoning [27]. Additionally, [127] highlights the need for explainability techniques that reveal prediction uncertainty and error boundaries to guide appropriate human-AI collaboration.

While the XLR-KGDD framework [128] advances clinician-oriented explanations, limitations like vague or inconsistent language in generated outputs exist. This would be improved by incorporating better guidelines and training on a larger medical corpus to produce more precise, context-aware explanations. This points to a broader need for adaptive and possibly interactive explanation systems that evolve with user needs and medical knowledge. Similarly, [127] emphasize that fixed, generic explanations may overwhelm users or fail to support trust calibration. Their findings stress the importance of designing explanations tailored to specific user groups and tasks, encouraging the development of adaptive, user-centric explainability approaches that address diverse goals such as trust calibration, safety, fairness, and user autonomy.

### 3) SCALABLE PRIVACY PRESERVATION

Despite advances in privacy-preserving methods, vulnerabilities remain, including latency issues and communication overheads [118]. Federated learning poses challenges in maintaining a balance between efficiency and privacy, with limited protection of participant identity [13]. Techniques like pseudonymization fall short due to re-identification risks. Future research must strengthen privacy guarantees through scalable, secure approaches suitable for real-world deployment [122].

### 4) ROBUSTNESS AND GENERALIZATION UNDER ADVERSARIAL CONDITION

AI systems must withstand various attacks (e.g., adversarial, closed box, poisoning) and adapt to real-world uncertainties. Key challenges include developing generalized attack strategies for robustness testing, generating adversarial examples efficiently, and mitigating robust overfitting. Hence, enhancing AI's ability to generalize under uncertainty remains critical. Similarly, OOD detection remains limited by the need for retraining, model redesign, or prior OOD knowledge-constraints impractical for pre-trained models [89]. Detecting near-distribution samples is especially difficult, as current detectors often fail in subtle shift scenarios. Research must develop robust, generalizable OOD detection strategies that operate without retraining and support broader trustworthiness goals like fairness and explainability.

## C. LLM-SPECIFIC CHALLENGES

LLMs pose unique challenges, notably hallucinations and lack of uncertainty expression. Addressing these requires improved human-labeled data and training methods to reduce overconfidence and enhance alignment [101]. The best

practices for constructing alignment datasets and understanding labeler influence remain open research problems. Emerging AI forms like agentic and embodied AI introduce additional safety considerations that warrant future exploration of the trustworthiness of these autonomous agentic AI systems [27].

While SPIN [82] shows strong performance in mitigating the trade-off between fairness and privacy, its applicability to other TAI dimensions remains underexplored. The method is specifically tailored to decouple fairness and privacy representations, but it is unclear whether the same approach can be generalized to conflicts involving other dimensions. For instance, since there also exist conflicts between fairness vs. robustness and explainability, does reducing the tradeoff between fairness and privacy invariably affect robustness and/or explainability? Future work is needed to assess whether similar forms of neuron coupling exist across these dimensions and if the suppression mechanism can be effectively adapted.

Challenges in explaining LLMs stem from their massive scale and unique behaviors. The sheer size of these models, with their billions of parameters and enormous training datasets, makes their internal workings incredibly complex and difficult to interpret. Traditional methods, such as gradient-based techniques or SHAP values, demand substantial computing power, making them impractical for use with modern LLMs. Also, a fundamental challenge is the lack of a ground truth for explanations. Since there is no definitive right answer for why an LLM made a specific choice, it is a challenge to design and evaluate the accuracy of new explanation algorithms. This lack of a reliable benchmark makes it hard to measure the faithfulness and fidelity of any proposed explanation method [108].

Robustness to increasingly complex adversarial contexts also demands further exploration. Beyond current jailbreak prompts, future studies should examine deceptive conversational setups (e.g., honeypot dialogues), evaluate resilience against coordinated adversarial agents, and develop sophisticated countermeasures tailored to such scenarios. Also, future research should investigate how objective trustworthiness metrics (e.g., toxicity, fairness, and ethics) correspond to human perceptions and values [104].

A promising direction for future work is to explore robust training strategies for the hallucination classifier to enhance its performance under distribution shifts between the unlabeled data used for training and the test data encountered during deployment. While [134] demonstrates strong generalization across datasets, its effectiveness can still degrade when there is a significant mismatch in domain, style, or content between training and inference scenarios. Developing techniques such as domain adaptation, distributionally robust optimization, or contrastive learning could help the classifier better distinguish truthful from hallucinated outputs even when the data distribution shifts. This would improve the reliability and applicability of hallucination detection methods in real-world, dynamic

environments where LLMs are deployed across diverse user inputs and evolving tasks.

D. AGENTIC AI

Agentic AI systems introduce new risks beyond those of traditional LLMs due to their autonomy, persistent memory, and dynamic decision-making. Their ability to make independent choices makes them complex, unpredictable, and difficult to secure. A primary concern is autonomy abuse, where agents may misinterpret user goals or pursue harmful strategies because of flawed reasoning. Their actions are not fixed, making it challenging to enforce safety measures. According to [31], persistent memory, while useful, is a unique vulnerability. Unlike stateless models, agents with memory can be compromised by subtle prompt injections that persist and spread harmful instructions. This issue is exacerbated by a lack of version control and auditing in many open-source frameworks. Other key threats include agent orchestration flaws, where a compromised or biased central controller can cause cascading failures and goal misalignment, as well as propagate discrimination which can lead to agents working against each other. This inadvertently results in algorithmic monoculture [146]. The ability of these agents to invoke external tools also introduces risks, such as DDoS attacks, as many systems lack adequate sandboxing.

Furthermore, multi-agent systems face the risk of collusion or behavioral drift. Agents designed to collaborate may reinforce each other’s errors, leading to groupthink and flawed outputs. These emergent behaviors are the most difficult to predict, as they arise from complex interactions between agents, memory, and tools. Real-world examples highlight these risks; for instance, LLM-based agents have demonstrated how persistent memory can lead to data leaks, and ChatDev showed how agents can get stuck in a self-reinforcing cycle of errors without external feedback [31].

Ensuring trust in Agentic AI requires balancing autonomous decision-making with explicit human oversight mechanisms. Embedding these systems within inclusive and socially accountable design frameworks, such as participatory design that integrates diverse cognitive, cultural, and accessibility needs, supports equitable deployment and mitigates risks of opaque autonomous actions [153].

Future research in Agentic AI must focus on improving adaptability, robustness, and security to meet the demands of increasingly complex and dynamic environments. Current models struggle to generalize across tasks or adapt efficiently without extensive retraining. To address this, research should explore meta-learning and transfer learning, enabling agents to leverage past experiences for rapid adaptation. Advancements in robust reinforcement learning are also essential to ensure agent reliability in uncertain or noisy conditions.

TABLE 13. AI applications in healthcare.

| Application Area                       | Contributions / Techniques                                                                                                                                                                                         |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Clinical Decision Support (CDS)        | Improves diagnostic accuracy; supports timely interventions.<br><i>Techniques:</i> LLM, NLP, RAG, Prompt Engineering, Fine-tuning [25], [84], [96].                                                                |
| Diagnostics & Predictive Analytics     | Reduces errors, improves patient outcomes.<br><i>Techniques:</i> CNN, RNN, Random Forest, NLP, XAI, Multi-Agent Systems [15], [68], [123].                                                                         |
| Medical Imaging Analysis               | Faster, more accurate image interpretation.<br><i>Techniques:</i> CNNs, Grad-CAM, SHAP [15], [25], [123].                                                                                                          |
| Disease-Specific Prediction            | Early warning for critical events.<br><i>Techniques:</i> RNNs, Time-series ML models, Multi-Agent Systems [25], [123].                                                                                             |
| Drug Discovery & Gene Editing          | Accelerates therapy development.<br><i>Techniques:</i> DL models, Reinforcement Learning, Knowledge Graphs [1], [123], [155].                                                                                      |
| Administrative Workflows               | Enhances operational efficiency.<br><i>Techniques:</i> LLMs, RPA, Knowledge Graphs, Multi-Agent Systems [15], [95], [123].                                                                                         |
| Patient Engagement / Conversational AI | Improves accessibility, patient satisfaction.<br><i>Techniques:</i> LLMs, RAG-based chatbots, NLP, Virtual Health Assistants [25], [95], [97].                                                                     |
| Personalized Medicine                  | Better long-term outcomes, reduced hospitalizations.<br><i>Techniques:</i> ML/DL, LLMs, NLP, Multi-Agent Systems, Federated Learning [15], [29], [99], [123].                                                      |
| Robotic Surgery                        | Enhances dexterity, reduces invasiveness.<br><i>Techniques:</i> Computer Vision, Reinforcement Learning [123].                                                                                                     |
| Genomics & Multi-omics                 | Advances precision medicine.<br><i>Techniques:</i> Multi-Agent Systems, Knowledge Graphs, DL [123].                                                                                                                |
| Mental Healthcare                      | Continuous, stigma-free support.<br><i>Techniques:</i> Chatbots, LLMs, RAG [123].                                                                                                                                  |
| Preventive Healthcare                  | Extends healthy lifespan, lowers costs.<br><i>Techniques:</i> Predictive ML models, Chatbots, Multi-Agent System [25], [98], [123].                                                                                |
| Disaster Management                    | Rapid, informed decision-making.<br><i>Techniques:</i> Multi-Agent Systems [99].                                                                                                                                   |
| Cross-disciplinary Research            | Accelerates scientific breakthroughs by aggregating and synthesizing data from diverse sources like genomics, proteomics, and clinical trials .<br><i>Techniques:</i> Multi-Agent Systems, Knowledge Graphs [123]. |

Note: CDS: Clinical Decision Support; CNN: Convolutional Neural Network; DL: Deep Learning; LLM: Large Language Model; ML: Machine Learning; NLP: Natural Language Processing; RAG: Retrieval-Augmented Generation; RNN: Recurrent Neural Network; RPA: Robotic Process Automation; XAI: Explainable AI.

Security remains a critical concern, as current defenses are vulnerable to evolving threats like hidden prompt injections and data poisoning. Future work should incorporate adversarial training and formal verification to

proactively strengthen the security posture of LLM-based agents.

#### E. ADDRESSING TRADE-OFFS AND INTERSECTIONALITY

TAI entails navigating trade-offs between competing dimensions such as fairness and performance or privacy and utility. Enhancing one may inadvertently undermine another, requiring a holistic, intersectional approach rather than isolated solutions. A deeper understanding of interdependencies is essential for crafting ethically sound trade-offs. As a result, [20] highlights that fairness in AI remains technically complex due to the absence of universally accepted definitions and methods. Conflicting fairness criteria and context-dependent sensitivity attributes necessitate tailored frameworks. Diverse stakeholder input is essential to address variations in fairness perceptions and to manage trade-offs among accuracy, fairness, and privacy. Moreover, conflicting principles like fairness and transparency demand nuanced, domain-specific resolutions. To demonstrate, consider the scenario in breast cancer screening, where an AI model is trained to detect tumors from mammograms. Fairness becomes technically complex because different stakeholders define it differently: radiologists, for instance may emphasize equal accuracy across age groups, whereas patient advocates might focus on reducing racial disparities in false negatives. At the same time, ensuring transparency in how the AI highlights suspicious regions may conflict with optimizing fairness constraints. Balancing these competing criteria requires a tailored framework that integrates input from clinicians, patients, ethicists, and regulators.

Conflict-aware multi-objective optimization offers a promising direction by training models to balance competing values according to specific domain risks [30]. For instance, relaxing fairness constraints may allow balanced outcomes with privacy, suitable for applications like healthcare triage. In contexts like emergency diagnostics, prioritizing robustness may be acceptable, whereas patient communication tools may emphasize fairness and transparency. Additionally, addressing these conflicts requires causal-dependency mapping, and scenario-based simulation to reason about trade-offs explicitly. Methodologies integrating anomaly detection with AI-enabled resilience, such as those used in digital twin technologies, offer a quantifiable view of intersectional trust and can help model the cascading effects of trade-offs across industrial and healthcare ecosystems [154].

#### F. INTEGRATION WITH REAL-WORLD SYSTEMS AND DATA MODALITIES

TAI research often focuses on structured tabular healthcare data, overlooking modalities like imaging, EHRs, and genomics. Evaluation methods must be tested in practical settings to bridge academic and clinical practice. Later ML lifecycle stages - testing, deployment, and monitoring, are underrepresented despite being essential for trustworthiness.

Data quality issues and multi-modality integration further complicate real-world deployment.

### V. CONCLUSION

#### A. SUMMARY

Artificial Intelligence (AI) holds immense promise in healthcare, from improving diagnostics and clinical workflows to enabling personalized care and accelerating drug discovery. However, progress has been hindered by concerns over AI's trustworthiness. This review examines the critical dimensions of Trustworthy AI (TAI), bias and fairness, privacy and security, robustness and safety, and explainability and transparency, highlighting their interdependencies and trade-offs. It emphasizes that AI's success in healthcare depends on establishing trust among patients, professionals, and regulators.

TAI encompasses not only technical reliability but also ethical and legal considerations. While TAI emphasizes transparency, accountability, and verifiability, there are overlaps with responsible AI in fairness, privacy, and organizational governance. Regulatory frameworks, including the EU's GDPR, the U.S.'s NIST and GAO models, and Canada's CPPA and now obsolete AIDA, reflect global efforts to operationalize TAI, particularly in health systems governed by laws like HIPAA. This review also considers the unique trust challenges posed by generative models, especially LLMs and Agentic AI, across alignment, reliability, privacy, and explainability dimensions.

#### B. KEY ISSUES

Despite significant developments, widespread adoption of TAI in healthcare faces persistent barriers. A major challenge is bias, which often stems from skewed training datasets and results in disparate model performance across demographic groups. The subjectivity of fairness definitions complicates the establishment of universal standards. Meanwhile, the opacity of AI systems (especially closed box models) undermines trust, particularly where explanation is vital for clinical validation. There is a need for human-centered explanations tailored to clinicians, patients, and policy makers.

Privacy and security remain critical challenges as AI models trained on sensitive health data are vulnerable to inference and extraction attacks. While privacy-preserving techniques show promise, they are not yet robust or scalable, particularly in federated learning contexts. LLMs and Agentic AI introduce further complexities in securing distributed data and maintaining identity protection. Similarly, robustness concerns persist, as AI systems must remain reliable under adversarial perturbations and unpredictable inputs. The inability to generalize to novel scenarios or distribution shifts without retraining continues to limit real-world deployment.

Organizational and human trust factors also present obstacles. Misalignment between actual system

trustworthiness and user perceptions can result in over-trust or under-trust, both of which compromise effective use. A lack of standardized, comprehensive frameworks for evaluating TAI, especially for qualitative aspects like human oversight, impairs meaningful cross-system comparisons. Finally, improving one TAI dimension often risks degrading another, such as when increasing privacy reduces explainability, illustrating the need for principled trade-offs.

### C. RECOMMENDATIONS

Addressing these challenges in healthcare requires coordinated efforts across technical, ethical, and regulatory domains. For example, data augmentation and inclusive data collection are critical when building diagnostic models, such as radiology AI systems that often underperform on scans from underrepresented populations. Similarly, post-deployment monitoring is essential in clinical settings like intensive care units, where patient demographics and disease prevalence shift over time, potentially introducing new inequities if left unchecked. Advanced techniques such as conflict-aware multi-objective optimization can help balance competing goals, ensuring that an AI system for predicting sepsis, for instance, maintains high diagnostic accuracy while safeguarding patient privacy and minimizing disparities across gender or ethnicity.

Explainability and transparency must become more human-centered, with methods offering actionable and intuitive insights. Objective metrics for explanation quality, robustness, consistency, interpretability, should guide development. There is a pressing need for explainable federated learning and logic-based methods that align technical transparency with practical interpretability for non-expert users.

Meanwhile, privacy and security enhancements must prioritize scalable, efficient defenses, particularly in federated environments and for LLMs. Continued research into techniques like differential privacy, secure aggregation, and resistance to inference attacks is essential to strengthen protection without degrading performance or usability. In addition, robustness and safety should be improved through the development of generalizable attack simulations and efficient adversarial training that prevents overfitting. More effective out-of-distribution detection strategies are needed to identify distribution shifts and reduce uncertainty in unpredictable clinical settings.

The development of agentic AI requires an approach centered on multi-dimensional safeguards, adaptability, and governance. The key recommendations include: establishing robust architectures with multi-agent designs for scalability and conflict resolution; implementing security-conscious development through adversarial testing and red-teaming; and adopting ethical governance to ensure transparency and interpretable decision-making. Hence, successful agentic AI must be built for human-agent collaboration and real-world

applications, integrating continuous oversight, accountability, and resilience to be both capable and trustworthy.

A holistic approach to TAI must include dynamic, automated evaluation frameworks that adapt to evolving threats and cover both quantitative and qualitative dimensions, including human oversight and societal impact. Equally important is integrating human values throughout AI development, ensuring systems are designed to complement, not replace, clinical judgment. For LLMs, future research should address hallucinations, refine alignment data, and develop techniques to express uncertainty and avoid overconfidence. Understanding how human feedback influences alignment remains an open and important question.

In conclusion, building truly trustworthy AI in healthcare demands sustained interdisciplinary collaboration, technical innovation, and adaptive regulation. By addressing the outlined challenges and embracing these recommendations, the healthcare community can realize AI systems that are not only accurate and efficient, but also ethically sound, transparent, and aligned with human values, fostering deep trust and transforming global health outcomes.

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